



Gyanmanjari[®]
Innovative University

Shaping Minds with **Skills** and **Innovations**



"Education Is Not The Learning Of Facts,
But The Training Of Mind To Think"

By Albert Einstein



Gyanmanjari[®]
Innovative University

COURSED OFFERED

DIPLOMA(DE)

CIVIL ENGINEERING | MECHANICAL ENGINEERING | INFORMATION TECHNOLOGY
ELECTRICAL ENGINEERING | COMPUTER ENGINEERING | CHEMICAL ENGINEERING
INTERIOR DESIGN | COMPUTER AIDED COSTUME DESIGN & DRESS MAKING
ELECTRONICS & COMMUNICATION

UNDER GRADUATION (B.TECH)

CIVIL ENGINEERING | MECHANICAL ENGINEERING | ELECTRICAL ENGINEERING
COMPUTER ENGINEERING | COMPUTER SCIENCE AND ENGINEERING
INFORMATION TECHNOLOGY | INFORMATION & COMMUNICATION TECHNOLOGY
CHEMICAL ENGINEERING | BIO-TECHNOLOGY | FOOD TECHNOLOGY
ELECTRONICS & COMMUNICATION | AERONAUTICAL ENGINEERING

UNDER GRADUATION (B.TECH) MINOR

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING | CYBER SECURITY
ADVANCE WEB DEVELOPMENT | ROBOTICS | CONSTRUCTION TECHNOLOGY
ELECTRIC VEHICLE | SYSTEM ENGINEERING | AQUA-FOOD TECHNOLOGY
DRUG ENGINEERING | INTERNET OF THINGS (IOT)

POST GRADUATION (M.TECH)

STRUCTURAL ENGINEERING | CONSTRUCTION AND PROJECT MANAGEMENT
MACHINE DESIGN | THERMAL ENGINEERING | MANUFACTURING TECHNOLOGY
ELECTRICAL ENGINEERING | ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
COMPUTER ENGINEERING | VLSI AND EMBEDDED SYSTEM
COMPUTER VISION AND IMAGE PROCESSING | CYBER SECURITY

ENGINEERING

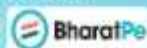
Why GMIU ?

Holistic development @ GMIU
(Gyanmanjari Group)

583+

COMPANY VISITED FOR PLACEMENT

HIGHEST

PLACEMENT PACKAGE OF
42 Lac. BY 

7+

STARTUPS

15+
Years

of Education
Experiences

10,000+

Alumni

100+

MOUs/Associating

200+

Facilities with
Ph.D./MCA/MBA/
MTech/NET/SLET/
GSET Qualified

46,000+

Books

200+
Gbps

Internet
Bandwidth

27+

Programmes

Dedicated

Departmental
Placement
Cell

60+

Laboratories

15+

IEP (International
Exchange Programs)

100+

Boys in
Hostel

100+

Girls In Hostel

**Pioneer
In**

Integrated
Dual Degree
Programs

**Pioneer
In**

Minor/Honour/
Specialization
Program

**Pioneer
In**

Higher Education
Parents Meet

**Pioneer
In**

Skill based
Education

**Pioneer
In**

Innovative
Placement Policy

**Pioneer
In**

Mobile APP
Moodle Based
Education

* Bhavnagar City Context

* Proposed.

Achievement @ GMIU



GMT Best Engineering College
Gujarat for Industry
Interface 2020



Gyanmanjari Institute of Technology
as the winner of ICI-Ultra Tech Outstanding
Structure Award 2020 (State)



Kahan Mehta
Computer Engineering,
Gold Medal in GTU



Gyanmanjari Institute of Technology
as the winner of ICI-Ultra Tech
Outstanding Structure
Award 2020 (State)

PLACEMENT ASSOCIATES



* & Many More...

PLACEMENT



OUR FACILITY



Android Lab



Transportation Facility



Box Cricket



IOS LAB



Class Room



Library

ACTIVITY & EVENT



Techmanjari @ GMIU



Celebrity @ GMIU



Traffic Awareness Rally



Sports @ GMIU



Social Activities @ GMIU



RELIANCE INDUSTRIAL VISIT



ADANI PORT INDUSTRIAL VISIT



Gyanmanjari[®]

Innovative University

ઘર બેઠા યુનીવર્સિટી કેમ્પસ ની મુલાકાત માટે અહીં ક્લિક કરો.

<https://bitly.ws/3heF6>



ઘર બેઠા એડમીશન ટોકન ફી ભરી કન્ફર્મ કરવા માટે અહીં ક્લિક કરો.

<https://bitly.ws/3gWA2>



FOR ADMISSION INQUIRY CALL



75 74 949494, 90 999 51160



Sidsar Road, Bhavnagar-364060,
Gujarat, India.



www.gmiu.edu.in



Std :10th (Eng. Med.)

Sub : ENGLISH

Date : 23-01-2025

Total Marks : 80

SECTION A

*** Read the following passage and select the appropriate answers from the options given below : (02)**

The creature that emerged from this sack on to the spacious tiled floor of the consulate bedroom resembled most of all a very small, mediocrally conceived, dragon. From the head to the tip of the tail he was coated with symmetrical pointed scales of mud armour, between whose tips was visible a soft velvet fur like that of a chocolate-brown mole. He shook himself, and I half expected a cloud of dust, but in fact it was not for another month that I managed to remove the last of the mud and see the otter, as it were in his true colours.

1. How did the author see the otter's true colour?
A. by turning him upside down. B. by lying him on the floor.
C. by removing the last of the mud. D. by touching the soft velvet fur.
2. Choose the synonym of the word 'covering' from the passage?
A. sack. B. armour C. mol D. symmetrical

*** Read the following passage and answer the questions : (03)**

The baker or bread-seller of those days had a peculiar dress known as the kabai. It was a single-piece long frock reaching down to the knees. In our childhood we saw bakers wearing a shirt and trousers which were shorter than full-length ones and longer than half pants. Even today, anyone who wears a half pant which reaches just below the knees invites the comment that he is dressed like a pader!

Questions:

3. What was the baker's dress known as?
4. Describe the baker's dress.
5. Who is compared with a pader in today's time?

*** Fill in the blanks choosing the correct words given in the bracket : (write only the answers.) (05)**
(courage, counted, sacrifices, repaid, suffering)

That day had come about through the unimaginable 6 of thousands of my people, people whose 7 and 8 can never be 9 or 10.

*** Answer any three of the following questions in five to six sentences each : (06)**

11. How was Mij to be transported to England?
12. How did the hailstones affect Lencho's field? What was Lencho's only hope?
13. Anne thought, 'Paper has more patience than people.' Justify.
14. Kisa Gotami again goes from house to house after she speaks with the Buddha. What does she ask for, the second time around? Does she get it? Why not?
15. How can you say that Coorg holds a special place in armed forces?

SECTION B

*** Read the following verse and answer the questions given below : (03)**

Or if some time when roaming round,
A noble wild beast greets you,
With black stripes on a yellow ground,
Just notice if he eats you.
This simple rule may help you learn
The Bengal Tiger to discern.

Questions:

16. According to the poet, how will the Bengal Tiger come to your notice?

17. From which poem has the given stanza been taken? Also mention the poet's name.
18. What does 'with black stripes on a yellow ground' describe?

*** Select the correct figures of speech from the options given below :**

(02)

19. 'In his quiet rage'.
A. Antithesis B. Oxymoron C. Personification D. Transferred Epithet
20. 'The trees are stumbling forward'.
A. Synecdoche B. Metaphor C. Personification D. Transferred Epithet

*** Answer any three of the following questions in five to six sentences each :**

(06)

21. Why does the poet hold with those who 'favour fire'?
22. According to 'The Ball Poem', what does "in the world of possession" mean?
23. Why is Amanda scolded for eating a chocolate?
24. How did Belinda's pets, other than Custard, face the pirate?
25. Why is the youth in despair in the poem 'For Anne Gregory'?

*** Read the following passage and answer the questions :**

(03)

There were about fifteen thousand pounds' worth of jewels in the Grange safe. If he sold them one by one, he expected to get at least five thousand, enough to make him happy for another year. There were three very interesting books coming up for sale in the autumn. Now he would get the money he wanted to buy them.

He had seen the housekeeper hang the key to the kitchen door on a hook outside. He put on a pair of gloves, took the key, and opened the door. He was always careful not to leave any fingerprints.

A small dog was lying in the kitchen. It stirred, made a noise, and moved its tail in a friendly way.

'All right, Sherry,' Horace said as he passed. All you had to do to keep dogs quiet was to call them by their right names, and show them love.

Questions:

26. Why did Horace require the money?
27. Why was he careful about not leaving any fingerprints?
28. According to Horace, how would the dog recognize him?

*** Read the following passage and answer the questions :**

(02)

Mme Loisel seemed old now. She had become a strong, hard woman, the crude woman of the poor household. Her hair badly dressed, her skirts away, her hands red, she spoke in a loud tone, and washed the floors with large pails of water. But sometimes, when her husband was at the office, she would seat herself before the window and think of that evening party of former times, of that ball where she was so beautiful and so flattered.

Questions:

29. Why did Mme Loisel look old?
30. What did Mme Loisel think about sometimes?

SECTION C

*** Rectify the errors in the given passage :**

(04)

	Error	Correction
31. Mij was out of a box.		
32. In a flash. He disappear.		
33. With high speed down the aircraft.		
34. There was squawks and shrieks.		

*** Punctuate the following passage appropriately : (Questions nos. 35 to 38)**

(02)

May we start now madam the conductor asked smiling.

50. Why was his action exceptionally courageous?

51. How did he lose his life?

52. Give the noun forms of : remember, excel **OR**

*** Read the following poem and answer the questions given below :**

(04)

"for oh," say the children, "we are weary"

And we cannot run or leap;

If we cared for any meadows sorely in the stooping,

We fall upon our faces, trying to go;

And underneath our heavy eyelids drooping

The reddest flower would look as pale as snow.

For, all day, we drag our burden tiring through the coal-dark, underground;

Or, all day we drive the wheels of iron.

In the factories, round and round."

- Elizabeth Barrett Browning

Questions:

49. What do you mean by the phrase, "reddest flower"?

50. How is the mood of the children in the poem?

51. Why cannot the children run or leap?

52. How does the work affect the children?

53. You recently visited an old age home. prepare a diary entry on the same.

4.

OR

53. You visited a bank to inquire about the procedure to open a savings account. Write a dialogue between you and the manager regarding the inquiry you made there.

4.

54. Showroom of Titan Watches is providing exclusive 40% off discount on selected watches. Draft an Advertisement for a local newspaper.

4.

OR

54. You are Shailja / Shikhar, the Sports Captain of Rotary Public School, New Delhi. Draft a notice informing the members of the School Cricket Team about a special coaching camp that is being organised in the school premises during the summer vacation.

4.

OR

54. Design a poster for your school library on the value of books and good reading habits, you may use slogans.

4.

SECTION E

55. A fire broke out in your locality which caused a lot of destruction. Write a report on this incident based on your observation.

4.

OR

55. Draft an e-mail to the Municipal Commissioner complaining about the hygiene standards of the nearby restaurants.

4.

OR

56. Your furniture has been damaged in transit by train. Write a letter to the Railways complaining about the Same and claiming damages.

6.

57. Write an essay in about 200 words on any one of the following :

(08)

(1) India – A Democratic Nation

(One of the biggest democracy – meaning – interest of the people – self – reliant after independence – Growth of sectors – problems for indian democracy – conclusion)

(2) If there were no Newspapers

(part of habit – variety of news – news and write-ups for all segments of society – knowing the world – powerful media)

OR

57. Write a story in about 200 words based on the inputs given below :

(08)

A fox – caught in trap – succeeded in getting away – had to leave his tail – kept away from other foxes – tried to make a plan – called a meeting of other foxes – told them about his trap – advised other foxes to cut off their tails – old fox gave him his answer – moral.

..BEST OF LUCK..

Std :10th (Eng. Med.)

Sub : Gujarati

Date : 22-01-2025

Total Marks : 80

વિભાગ - A ગદ્ય વિભાગ

(અ) નીચેનાં જોડકાં યોગ્ય રીતે જોડીને ફરીથી લખો:

[03]

‘અ’ (કૃતિ)	‘બ’ (સાહિત્યપ્રકાર)
1. શરણાર્થના સૂર	(અ) લોકકથા
2. સૂરજ તો બધે સરખો	(બ) ટૂંકી વાર્તા
3. જેઠીબાઈ	(ક) રેખાચિત્ર
	(ડ) હાસ્યનિબંધ

(બ) (1) નીચે આપેલા વિકલ્પોમાંથી યોગ્ય વિકલ્પ પસંદ કરી ખાલી જગ્યા પૂરો :

[02]

- કાન્તિ..... ના ખાંચામાં સંતાઈને બેઠો. (લાલજી મોચી, હરજીવન મિસ્ત્રી, કંચનમાસી)
- ‘પ્રાણનો મિત્ર’ વાર્તા આપણને..... માવજત માટે જાગૃત કરે છે. (પ્રાણીની, પર્યાવરણની, તંદુરસ્તીની)

(2) નીચે આપેલા વિધાનો ખરાં છે કે ખોટાં તે જણાવો:

[02]

- રમઝુ મીરની પુત્રીનું નામ ‘ગવરી’ હતું.
- ‘ટિકિન’ કૃતિના લેખકનું નામ ‘પ્રેમજી પટેલ’ છે.

(ક) નીચે આપેલા પ્રશ્નોના એક વાક્યમાં ઉત્તર લખો :

[03]

- શંભુ પુરાણીનો ભાણેજ વલભો મુંબઈ જઈ શું ભણી આવ્યો હતો?
- ‘સર અમલ દેરાસરી’ શું મેળવીને રાજી હતા?
- ગુજરાતના સુલતાનોના સમયમાં દીવ કોણે જીતી લીધું હતું?

(ડ) નીચે આપેલા પ્રશ્નોના બે-ત્રણ વાક્યમાં ઉત્તર લખો : (ગમે તે બે)

[04]

- ‘જર્મની’ વિશે લેખકને શો ખ્યાલ હતો?
- અંબારામ ભદ્રંભદ્રના વેગ માટે કઈ ઘટના યાદ કરાવે છે?
- બા જ્યારે હિંમતને તેડતી ત્યારે કાન્તિના દિલમાં કેવો ભાવ જાગતો?

(ઈ) નીચે આપેલા પ્રશ્નોના સવિસ્તર વાક્યોમાં ઉત્તર લખો : (ગમે તે બે)

[06]

- ભૂલ કરનારાએ દંડની ટિકિટની ફોટોસ્ટેટ નકલ કરી ઘરની દીવાલ પર કેમ રાખી?
- ‘દીપક’નું પાત્રાલેખન કરો.
- ‘પારવતીબા’ અને ‘મોટા ભાઈ’ માટે રમાના મનમાં કેવો ઈર્ષ્યાભાવ હતો તે ‘નથી’ પાઠને આધારે સમજાવો.

વિભાગ – B પદ્ય વિભાગ

(અ) નીચેનાં જોડકાં યોગ્ય રીતે જોડીને ફરીથી લખો:

[03]

‘અ’ (કૃતિ)	‘બ’ (સાહિત્યપ્રકાર)
17. મોરલી	(અ) લોકગીત
18. તે બેસે અહીં	(બ) ઊર્મિકાવ્ય
19. હાથ મેળવીએ	(ક) ગઝલ
	(ડ) પદ

(બ) (1) નીચે આપેલા વિકલ્પોમાંથી યોગ્ય વિકલ્પ પસંદ કરી ખાલી જગ્યા પૂરો :

[02]

20. સુશોભનથી પ્રસન્ન થાય છે. (મન, દીવાનખાનું, તન)
 21. કવિનો નો દીપક કદી ન ઓલવાય એમ ઈચ્છે છે. (અંધશ્રદ્ધા, શ્રદ્ધા, નિરાશા)

(2) નીચે આપેલા વિધાનો ખરાં છે કે ખોટાં તે જણાવો:

[02]

22. ‘હાઈકુ’ ગુજરાતી ભાષાનો કાવ્યપ્રકાર છે.
 23. ‘તે બેસે અહીં’ કાવ્ય ‘યદા તદા ગઝલ’ સંગ્રહમાંથી લેવામાં આવ્યું છે.

(ક) નીચે આપેલા પ્રશ્નોના એક વાક્યોમાં ઉત્તર લખો :

[03]

24. પંખીએ એક માણસ સામે જોયા પછી શું કર્યું?
 25. ‘હાથમાં ઉષ્મા અને થડકો’ કહીને કવિ કયા ભાવને ભેળવવાની વાત કરે છે?
 26. ‘જીવન અંજલિ થાજો’ કાવ્યના કવિ ભૂખ્યા અને તરસ્યા માટે શું બનવાનું ઈચ્છે છે?

(ડ) નીચે આપેલા પ્રશ્નોના બે-ત્રણ વાક્યોમાં ઉત્તર લખો : (ગમે તે બે)

[04]

27. ‘મોરલી’ કાવ્યના આધારે કૃષ્ણના પહેરેવેશનું વર્ણન કરો.
 28. ‘દીવાનખાનામાં’ કાવ્યના અંતે નાયિકા કેવો મનોભાવ અનુભવે છે?
 29. તળાવને કોણ જગાડે છે.

(ઈ) નીચે આપેલા પ્રશ્નોના સવિસ્તર વાક્યોમાં ઉત્તર લખો : (ગમે તે બે)

[06]

30. એવો કયો કાવ્યપ્રકાર છે જેમાં ‘રદીક’ અને ‘કાફિયા’ મુખ્ય અંગો છે અને જેમાં હૃદયને સ્પર્શે તેવા ભાવ વણાયેલા હોય છે? તમારા પાઠ્યપુસ્તકમાં આવી કઈ રચના છે? તેમાંથી આપણને શું શીખવા મળે છે?
 31. ‘જીવન અંજલિ થાજો’ – શીર્ષકની યથાર્થતા સિદ્ધ કરો.
 32. ‘ક્યાં રે વાગી’ કાવ્યનો ભાવાર્થ સમજાવો.

વિભાગ – C વ્યાકરણ વિભાગ

> માગ્યા પ્રમાણે પ્રશ્નોના ઉત્તર લખો:

33. નીચેનામાંથી કયો શબ્દ ‘મિત્રતા’ શબ્દનો સમાનાર્થી નથી.

[01]

- (A) વય (B) વખાણ (C) ભાઈબંધી

34. નીચેનામાંથી સમાનાર્થી શબ્દનું કયું જોડકું ઉચિત નથી ?

[01]

- (A) પારિતોષિક – ઈનામ (B) ઉર – નેણ (C) મતિ – મેઘા

35. ‘દયાળુ’ શબ્દનો વિરુદ્ધાર્થી શબ્દ શોધીને લખો.

[01]

- (A) વિષાદ (B) સક્રિય (C) નિષ્ક્રુર

36. નીચે વિરુદ્ધાર્થી શબ્દનાં જોડકાં આપ્યાં છે, તેમાં કયું જોડકું ઉચિત છે.

[01]

- (A) નિર્બળ × સબળ (B) પાપ × પુણ્ય (C) હરખ × આનંદ

37. નીચેનામાંથી કઈ જોડણી સાચી છે? [01]
 (A) રેટીયો (B) રેડિયો (C) રેટીયો
38. વગડા વચ્ચે આસુદેવ મહાદેવનું મંદિર છે. (યોગ્ય જોડણી વડે ખાલી જગ્યા પૂરો.) [01]
 (A) એકાકિ (B) એકાકી (C) એકાંકી
39. નીચેનામાંથી 'પાવક' શબ્દની કઈ સંધિ યોગ્ય રીતે છોડેલી છે? [01]
 (A) પો + અક (B) પા + અક (C) પા + અક
40. નીચેનામાંથી કયો સમાસ દ્વંદ્વ છે? [01]
 (A) રાતદિવસ (B) વ્યવહારડાહ્યાં (C) હાથરમાલ
41. 'ડગળી ચસકી જવી'.(રૂઢિપ્રયોગનો અર્થ આપો) [01]
42. 'ઘોબીનો કૂતરો, ન ઘરનો ન ઘાટનો'.(કહેવત સમજાવો) [01]
43. 'દીપક વડે ફરવા જવાય છે'.(કર્તરિવાક્ય બનાવો) [01]
44. 'કંચન માસી બોલ્યા'.(ભાવેવાક્ય બનાવો) [01]
45. સત્તરમા સૈકાની અઘવચ્ચેનો એ સમય હતો. (રેખાંકિત વિશેષણનો પ્રકાર જણાવો) [01]
46. નીચેનામાંથી વિકારી વિશેષણ શોધો. [01]
 (A) નખરાળું (B) રેશમી (C) એકાકી
47. દરેક જણ અધું જ ખાઈ જાય કિશમાં એક દાણોય છાંડે નહિ (યોગ્ય વિરામચિહ્નો મૂકી વાક્ય ફરીથી લખો) [01]
48. સામાન્ય રીતે કોઈ ઉક્તિ, સંવાદ દર્શાવવા હોય તો કયું વિરામચિહ્ન વાપરી શકાય? [01]
 (A) અલ્પવિરામ (B) બેવડાં અવતરણ (C) એકવડાં અવતરણ

વિભાગ - D અર્થગ્રહણ / લેખન વિભાગ

49. નીચે આપેલા ગદ્યખંડનો આશરે ત્રીજા ભાગમાં સંક્ષેપ કરી તેને યોગ્ય શીર્ષક આપો. [04]
 “કુદરત માણસને જીવન બક્ષે છે, એ જીવનને શુદ્ધ, સમૃદ્ધ, પ્રસન્ન, ગંભીર અને વૈવિધ્યપૂર્ણ બનાવવાના પ્રયત્નમાં માણસે સંસ્કૃતિ ખીલવી. પથ્થરમાંથી જેમ મૂર્તિ બનાવીએ છીએ, ધ્વનિમાંથી સંગીત ઉપજાવીએ છીએ, ઘર્ષણમાંથી જેમ જવાળા સળગાવીએ છીએ, તેમ જ જીવનમાંથી સંસ્કૃતિ ખીલવીએ છીએ. જીવન એ પ્રકૃતિ છે. સંસ્કૃતિ એ એનો ઓપ છે. જીવન જો સંસ્કારિતાનો ઓપ ધારણ ન કરે તો એ પરિપૂષ્ટ હોવા છતાં, કૃતાર્થ ન થઈ શકે; અને સંસ્કૃતિ જો જીવનને વફાદાર ન રહે તો એમાં પોલાપણું આવે એટલું જ નહિ, પણ વિકૃતિ પેદા થઈને દુર્ગંધ જ છૂટે. જો જીવન ઘરતી હોય તો સંસ્કૃતિ એનું સ્વર્ગ છે. એ બે વચ્ચે જો અનુસંધાન ન સધાય તો બંને વ્યર્થ થઈ જાય.”
50. આપેલા મુદ્દા પરથી વાર્તા લખો. તેમાંથી મળતો બોધ લખો, તેને યોગ્ય શીર્ષક આપો: [04]
મુદ્દા :- એક મુસાફરનું ચાલતાં-ચાલતાં થાકી જવું - થાક દૂર કરવા આંબાના ઝાડ નીચે આડા પડવું - વિચારવું : ‘ઈશ્વર મૂર્ખ છે, વેલ નાની અને એની ઉપર કોળા જવું મોટું ફળ! જ્યારે આંબાનું વૃક્ષ મોટું અને એની ઉપર કેરી જેવું નાનું ફળ!’ - ઠંડી હવા - ઊંઘ આવી જવી - એક કેરી તૂટીને માથા પર પડવી - ઊંઘ ઊડી જવી - માથામાં દુખાવો - ઈશ્વરી યોજનાનું રહસ્ય સમજવું - ‘ઈશ્વર બુદ્ધિમાન છે’ - બોધ
51. તમારી શાળામાં યોજાયેલા ‘રમતોત્સવ’ નો અહેવાલ આશરે 100 શબ્દોમાં લખો. [04]
52. નીચે આપેલી પંક્તિઓનો વિચાર - વિસ્તાર કરો. [04]
 સફળતા જિંદગીની હસ્તરેખામાં નથી હોતી; **અથવા**
 ચણાયેલી ઈમારત તેના નકશામાં નથી હોતી.
 આચાર એ જ પ્રચારનું શ્રેષ્ઠ માધ્યમ છે.

• વૃક્ષો આપણા મિત્રો

મુદ્દા :- પ્રસ્તાવના - વૃક્ષો માનવીના પરમ મિત્રો - પર્યાવરણની જાળવણી - વૃક્ષો આપણને વિવિધ રીતે ઉપયોગી - વૃક્ષોથી થતા લાભ - પર્યાવરણને બચાવવાના ઉપાયો - ઉપસંહાર

અથવા

• કુદરતનાં હાસ્ય અને તાંડવ

મુદ્દા :- પ્રસ્તાવના - કુદરતનાં વિવિધ સ્વરૂપો - કુદરતનાં હાસ્ય સ્વરૂપો - કુદરતનાં તાંડવ સ્વરૂપો - ન્યાયી કુદરત દ્વારા સમાનતા - કુદરત આગળ માનવીની લાચારી - ઉપસંહાર

અથવા

• કન્યા - કેળવણીનું મહત્વ

મુદ્દા :- પ્રસ્તાવના - કન્યા-કેળવણી દ્વારા સમાજની સુધારણા - વર્તમાન યુગમાં સ્ત્રીની ભૂમિકા - પુરુષ સમોવડી નારી - ઉપસંહાર

..

Std :10th (Eng. Med.)

Sub : Gujarati

Date : 25-01-2025

Total Marks : 80

વિભાગ - A ગદ્ય વિભાગ

(અ) નીચેનાં જોડકાં યોગ્ય રીતે જોડીને ફરીથી લખો:

[03]

‘અ’ (પાત્ર)	‘બ’ (કૃતિ)
1. અમથીમા	(અ) ઝબકજયોત
2. મેનાબહેન	(બ) શરણાઈના સૂર
3. ગવરી	(ક) નથી
	(ડ) ટિફિન

(બ) (1) નીચે આપેલા વિકલ્પોમાંથી યોગ્ય વિકલ્પ પસંદ કરી ખાલી જગ્યા પૂરો :

[02]

4. સંભારણું રમઝુ માટે જરાય સુખદ નહોતું. (પત્નીનું, ગવરીનું, સકીનાનું)
5. ‘સહકારી મંડળીની સ્થાપના અંગેના વિચારબીજનું વાવેતર કરનાર હતા.
(ડો.વર્ગીસ કુરિયન, સરદાર વલ્લભભાઈ પટેલ, ત્રિભુવનદાસ પટેલ)

(2) નીચે આપેલા વિધાનો ખરાં છે કે ખોટાં તે જણાવો:

[02]

6. વર્ગીસ કુરિયનની આત્મકથા મારું સ્વપ્ન છે.
7. રમઝુએ મૃત પત્નીની કબર પાસે બેસી છેલ્લી સુરાવટ છેડી નહિ.

(ડ) નીચે આપેલા પ્રશ્નોના એક વાક્યમાં ઉત્તર લખો :

[03]

8. ‘બલાઈ’નું પાત્ર તમારા પુસ્તકમાં કઈ કૃતિમાં છે?
9. ભૂલ કરનારા જર્મનીની કઈ જગ્યાએ રેસ્ટોરામાં ગયા?
10. ‘નથી’ કૃતિમાં પારવતીબાના મોટા પુત્રનું નામ જણાવો.

(ડ) નીચે આપેલા પ્રશ્નોના બે-ત્રણ વાક્યમાં ઉત્તર લખો : (ગમે તે બે)

[04]

11. પારવતીબાનું ઘર કેવું હતું?
12. અરીસામાં પોતાનું મોઢું જોયા પછી દીપક કઈ કલ્પના કરે છે?
13. પોર્ટુગીઝ ઓફિસરો જેઠીબાઈના ઘર પાસેથી પસાર થતા ત્યારે શું કરતાં?

(ઈ) નીચે આપેલા પ્રશ્નોના સવિસ્તર વાક્યોમાં ઉત્તર લખો : (ગમે તે બે)

[06]

14. જેઠીબાઈનું પાત્રાલેખન કરો.
15. ‘ટિફિન’ કૃતિનો સારાંશ સમજાવો.
16. ‘શરણાઈના સૂર’ કૃતિને આધારે પિતાની વેદનાનું વર્ણન કરો.

વિભાગ – B પદ્ય વિભાગ

(અ) નીચેનાં જોડકાં યોગ્ય રીતે જોડીને ફરીથી લખો:

[03]

‘અ’ (કૃતિ)	‘બ’ (સાહિત્યપ્રકાર)
17. દીવાનખાનામાં	(અ) લોકગીત
18. ક્યાં રે વાગી	(બ) હાઈકુ
19. પર્વત ટોચે	(ક) અછાંદસ
	(ડ) મુક્તક

(બ) (1) નીચે આપેલા વિકલ્પોમાંથી યોગ્ય વિકલ્પ પસંદ કરી ખાલી જગ્યા પૂરો :

[02]

20. કાનાની મોરલીએ ઘેલો બન્યો. (ગોવાળ,ગરબો,ગોપીઓ)
 21. કાવ્યનાયિકા વસ્તુઓ વ્યવસ્થિત કરી રહ્યા છે. (રસોડામાં,દીવાનખાનામાં,ઓસરીમાં)

(2) નીચે આપેલા વિધાનો ખરાં છે કે ખોટાં તે જણાવો:

[02]

22. સોનેટ ચૌદ પંક્તિનું હોય છે.
 23. “હરિના લોચનિયા” પુસ્તકમાંથી ‘જીવન અંજલિ થાજો’ લેવામાં આવી નથી.

(ક) નીચે આપેલા પ્રશ્નોના એક વાક્યોમાં ઉત્તર લખો :

[03]

24. કેવો ઘડો છલકાય છે?
 25. મીરાંબાઈનો જન્મ ક્યાં થયો હતો?
 26. બધું ગોઠવાઈ ગયા પછી ‘દીવાનખાનામાં’નાં નાયિકા શું શોધે છે?

(ડ) નીચે આપેલા પ્રશ્નોના બે-ત્રણ વાક્યોમાં ઉત્તર લખો : (ગમે તે બે)

[04]

27. ‘જીવન અંજલિ થાજો’ કાવ્યના કવિ ક્યાં ચાલવાનું કહે છે?શા-માટે?
 28. વરસાદ વરસવાથી પ્રકૃતિ પર તેની શી અસર થઈ?
 29. તમારા હાથમાં શું શું છે.

(ઈ) નીચે આપેલા પ્રશ્નોના સવિસ્તર વાક્યોમાં ઉત્તર લખો : (ગમે તે બે)

[06]

30. પ્રથમ વરસાદ થયા પછીની નવચેતનાને તમારા શબ્દોમાં લખો.
 31. ‘મોરલી’ કાવ્યને આધારે કૃષ્ણની રાસલીલાનું વર્ણન કરો.
 32. “વાદળ ટોચે પહોંચી વેરાઈ ગયું” પંક્તિમાં વ્યક્ત થતો સંદેશ લખો.

વિભાગ – C વ્યાકરણ વિભાગ

➤ માગ્યા પ્રમાણે પ્રશ્નોના ઉત્તર લખો:

33. નીચેનામાંથી કયો શબ્દ ‘વ્યથિત’ શબ્દનો સમાનાર્થી નથી.
 (A) દુઃખિત (B) પીડિત (C) વિચલિત
 34. નીચેનામાંથી સમાનાર્થી શબ્દનું કયું જોડકું ઉચિત નથી ?
 (A) ઉચાટ – ફિકર (B) તેજ – પ્રકાશ (C) પથ્થર – તિમિર
 35. ‘પરિચિત’ શબ્દનો વિરુદ્ધાર્થી શબ્દ શોધીને લખો.
 (A) અપરિચિત (B) અન્યાય (C) અપરિણય
 36. નીચે વિરુદ્ધાર્થી શબ્દનાં જોડકાં આપ્યાં છે, તેમાં કયું જોડકું ઉચિત છે.
 (A) આજ × સાજ (B) ઈચ્છા × અનિચ્છા (C) હાડમારી × આફત
 37. નીચેનામાંથી કઈ જોડણી સાચી છે?
 (A) શિખરિણી (B) શિખરીણી (C) શીખરીણી

[01]

[01]

[01]

[01]

[01]

38. કાન્તિ ઘેર આવતો ત્યારે બા તેનાં લેતી. (યોગ્ય જોડણી વડે ખાલી જગ્યા પૂરો.) [01]
 (A) કપડાં (B) પૈસા (C) દુઃખણાં
39. નીચેનામાંથી 'સ્વાર્થ' શબ્દની કઈ સંધિ યોગ્ય રીતે છોડેલી છે તે શોધો: [01]
 (A) સ્વ + અર્થ (B) સ્ + અર્થ (C) સ્વા + ર્થ
40. નીચેનામાંથી કયો સમાસ દ્વન્દ્વ સમાસ છે? [01]
 (A) પીતાંબર (B) નર્મદા (C) જન્મ-મૃત્યુ
41. 'દીપક બુઝાવો'. (રૂઢિપ્રયોગનો અર્થ આપો) [01]
42. 'પાઠ ભણાવવો'. (કહેવત સમજાવો) [01]
43. દીપક ફરવા જાય છે. (કર્મણિવાક્ય બનાવો) [01]
44. રમઝુ બારણું બંધ કરી ગયો. (ભાવેવાક્ય બનાવો) [01]
45. ત્રણ-ત્રણ વાર દાદ આપવા છતાં રમઝુએ શરણાઈ બંધ કરી નહિ. (રેખાંકિત વિશેષણનો પ્રકાર જણાવો) [01]
46. નીચેનામાંથી અવિકારી વિશેષણ શોધો. [01]
 (A) ખરાબ (B) મોટું (C) ઝીણું
47. બરફનો ગાંગડો લાવ્યો છે કે ગાંગડી (યોગ્ય વિરામચિહ્ન મૂકી વાક્ય ફરીથી લખો) [01]
48. વાક્યને અંતે કયું વિરામચિહ્ન મૂકાય? [01]
 (A) અલ્પવિરામ (B) પૂર્ણવિરામ (C) ગુરૂવિરામ

વિભાગ - D અર્થગ્રહણ / લેખન વિભાગ

49. નીચે આપેલા ગદ્યખંડનો આશરે ત્રીજા ભાગમાં સંક્ષેપ કરી તેને યોગ્ય શીર્ષક આપો. [04]
 પશ્ચાતાપ એ એક કીમતી ધર્માનુભવ છે. જે વાસણ માટી કે રાખથી સાફ થતાં નથી તે જેમ તેજાબથી સાફ થાય છે તે કાટ બધો ઓગાળી કે બળી જઈ વાસણ ચળકવા માંડે છે, તેવી રીતે તેની ધર્મજીવન પર બાઝેલી કેટલીયે શિથિલતા પશ્ચાતાપથી દૂર થાય છે. અને માણસ એકાએક પલટો ખાય છે. ખરેખર પશ્ચાતાપની અસર તેજાબથી ઓછી નથી હોતી. સૃષ્ટિ માટે ને એમાંય માનવજીવન માટે ઈશ્વરની યોજનામાં મંગળ ને પવિત્ર, એમાં ભંગ પડી શકે માનવીની મૂર્ખાઈથી, પણ એ ભંગાણ પાછું સંધાય છે. માનવીના પશ્ચાતાપથી ને ઈશ્વરની અનહદ કૃપાથી. આખરે એની એ દિવ્ય યોજના આગળ ચાલવાની જ. – કાકા કાલેલકર

50. આપેલા મુદ્દા પરથી વાર્તા લખો. તેમાંથી મળતો બોધ લખો, તેને યોગ્ય શીર્ષક આપો: [04]
મુદ્દા :- ધ્રુવ અને ધર્મરાજની સારી મિત્રતા – ધ્રુવ – ધર્મરાજની પ્રકૃતિ, સ્વભાવ જુદાં જુદાં – ધર્મરાજની મોટી બહેન ધ્યાની ખૂબ જ રમતિયાળ – ધ્રુવની ઉપેક્ષા ધ્યાની માટે – રસ્તામાં ધ્રુવનું અકસ્માત – સમયસર ધ્યાનીનું આગમન – ધ્રુવનો બચાવ – ધ્યાનીનો છેવટે બહેન તરીકે સ્વીકાર – બોધ

51. તમારી શાળામાં 'સ્વચ્છતા અભિયાન' ઊજવવામાં આવ્યું હતું તે પ્રસંગનો અહેવાલ આશરે 100 શબ્દોમાં લખો. [04]

52. નીચે આપેલી પંક્તિઓનો વિચાર - વિસ્તાર કરો. [04]

મને એ જોઈને હસવું હજારો વાર આવે છે,
 પ્રભુ તારા બનાવેલા તને આજે બનાવે છે.

અથવા

વરસ્યો જઈને વાડ, મેઢુલિયો શા કામનો,
 મોટો લાગ્યો તાડ, ઢાંકે નહિ નિજ પંડને.

સંતોષી નર સદા સુખી

મુદ્દા :- શરૂઆત - સંતોષ એટલે શું? - સંતોષના લાભ - અસંતોષના ગેરલાભ - સંતોષની મર્યાદા - વિશાળ તૃષ્ણા
અથવા

• ગ્રીષ્મનો મધ્યાહન

મુદ્દા :- મધ્યાહનનું વાતાવરણ - મધ્યાહનનું મૌન - બપોરનું સૌંદર્ય - પંખી અને માનવીની હાલત - ગરમીથી બચવાના
ઉપાયો - સમાપન

અથવા

• ધોધમાર વરસાદે સર્જેલા દ્રશ્યો

મુદ્દા :- શરૂઆત વધુ વરસાદથી રચાતું વાતાવરણ - પશુ પક્ષીઓની અવદશા - શહેરોમાં ઊભી થતી સમસ્યાઓ
- જનજીવન તથા ખેતી પર થતી માઠી અસરો - નદીઓના ઘોડાપૂર - ભયાનક દ્રશ્યો - કરુણતા - રોગચાળો
- રાહત કાર્યો - વિવિધ સંસ્થાઓનો ફાળો - સરકારની મદદ - ગરીબોની ખરાબ દશા - માનવતા - સમાપન

..BEST OF LUCK..

Date: - 18/01/2025
Std: - 10th Eng. Med.

MATHEMATICS (BASIC)

Time: 3 Hours
Total Mark: - 80

Instructions:

1. In all, **54** questions in this question paper are divided into **four** sections A, B, C and D.
2. The numbers on the right side represents the marks of that section.
3. Draw figure, wherever necessary. Maintain the lines and arcs of the construction.
4. Start writing a new section from a new page. Answer the questions in serial order.
5. Use of calculator is strictly prohibited.

SECTION-A

(Question 1 to 24 : 1 Mark each)

Total : 24 Marks

➤ Choose the correct option in the following :

1. If the pair of equations $2x + 7y = 13$ and $5x + ky = 32$ has a unique solution, then holds good.
(A) $k = 17.5$ (B) $k = 15.7$ (C) $k \neq 17.5$ (D) $k \neq 15.7$
2. If $\frac{5}{2}$ is one of the roots of the equation $10x^2 - 29x + k = 0$, then $k = \dots\dots\dots$
(A) 25 (B) -25 (C) -10 (D) 10
3. For a given AP, if $d = -4$, $n = 7$ and $a_n = 4$, then $a = \dots\dots\dots$
(A) 6 (B) 7 (C) 20 (D) 28
4. Point $(5, -12)$ lies on a circle with the origin as the centre. Then, the radius of that circle is..... units.
(A) -7 (B) 7 (C) 13 (D) 10
5. $\cos^4\theta - \sin^4\theta = \dots\dots\dots$
(A) 1 (B) 0 (C) $\cos^2\theta - \sin^2\theta$ (D) $\cos^2\theta + \sin^2\theta$
6. For some given data, if $Z + \bar{x} = 98$ and $Z - \bar{x} = 12$, then $M = \dots\dots\dots$ by the inter relationship of mean, median and mode.
(A) 55 (B) 43 (C) 47 (D) 45

➤ **Fill in the blanks to make the statements correct.**

7. If $\text{HCF}(65, 117) = 3k - 2$, then $k = \dots\dots\dots$ (5, 4, 3)
8. If the product of the zeroes of the polynomial $p(x) = 6x^2 - x + k$ is $-\frac{1}{3}$, then $k = \dots\dots\dots$ (2, -2, 6)
9. In a 80-mark test of Mathematics, the probability of Rayna scoring 80 out of 80 marks is $\dots\dots\dots \left(\frac{1}{80}, 1, \frac{1}{81}\right)$
10. $\tan^2 \theta - \sec^2 \theta = \dots\dots\dots$ (0, 1, -1)
11. Point P lies in the exterior of a circle with centre O. Tangents PA and PB drawn from point P are inclined to each other at 58° . Then $\angle POA = \dots\dots\dots$ ($122^\circ, 58^\circ, 61^\circ$)
12. For a given frequency distribution, if $\sum f_i x_i = 245$ and $\sum f_i = 100$, then $\bar{x} = \dots\dots\dots$ (24.5, 2.45, 245)

➤ **State whether the following are true or false?**

13. The LCM of 35 and 42 is 35×42 .
14. For the quadratic polynomial $p(x) = x^2 + 5x + 4$, the sum of the zeroes is greater than the product of the zeroes.
15. The pair of linear equations $5x - 15y = 8$ and $3x - 9y = \frac{24}{5}$ has infinitely many solutions.
16. If $P(A) = \frac{3}{4}$, then $P(\bar{A}) = \frac{4}{3}$.

➤ **Answer the following in one word, sentence, number.**

17. Find the common difference of the AP $-5, -1, 3, 7, \dots$
18. At the most how many tangents can be drawn parallel to the diameter of a circle?
19. A letter is chosen at random from the English alphabet. Find the probability that the chosen letter is a vowel.
20. If the median of the observations 8, 12, 17, x , 25, 28 is 20, find x .

➤ **Match the following pairs correctly :**

'A'	'B'
21. LSA of a cuboid	(a) Perimeter of base $\times$ Height
22. Volume of a cone	(b) Area of base $\times$ Height
	(c) $\frac{1}{3} \times$ Area of base $\times$ Height

'A'	'B'
23. Sector	(a) Portion of a circular region enclosed by an arc and two radii.
24. Segment	(b) Any part of the circumference of a circle.
	(c) Portion of a circular region bounded by an arc and the end-points of the arc.

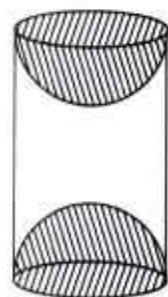
SECTION-B

Answer any 9 out of the following 13 questions with calculations.

(Question 25 to 37 : 2 Marks each)

Total : 18 Marks

25. Find the zeroes of the quadratic polynomial $3x^2 - x - 4$.
26. Find a quadratic polynomial for which the sum and the product of zeroes are 0 and $\sqrt{5}$ respectively.
27. Find the roots of the equation $2x^2 - 5x + 3 = 0$ by, factorisation.
28. For a given AP, if $a = 7$, $a_{13} = 35$, then find d and S_{13} .
29. Find the 20th term from the last term of the AP 3, 8, 13, ..., 253.
30. Find the values of y for which the distance between the points $P(2, -3)$ and $Q(10, y)$ is 10 units.
31. If $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$ are the vertices of a parallelogram taken in order, find x and y .
32. In $\triangle ABC$, right angled at B , $AB = 24$ cm, $BC = 7$ cm. Determine $\sin C$ and $\tan C$.
33. Evaluate : $\frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$.
34. A pole stands erect on the ground. From a point on the ground 20 m away from the base of the pole, the angle of elevation of the top of the pole is found to be 60° . Find the height of the pole. ($\sqrt{3} = 1.73$)
35. A wooden article was made by scooping out a hemisphere from each end of a solid cylinder, as shown in the given figure. If the height of the cylinder is 10 cm, and its base is of radius 3.5 m, find the total surface area of the article.



36. The largest sphere is carved out of a cube of the side 7 cm. Find the volume of the sphere.
37. For a given frequency distribution, in usual notations, $l = 125$, $n = 68$, $cf = 22$, $f = 20$ and $h = 20$. Find the median of the data.

SECTION-C

Answer any 6 out of the following 9 questions with calculations.
(Question 38 to 46 : 3 Marks each)

Total : 18 Marks

38. Solve the following pair of linear equations by substitution method :
 $2x + 3y = 11$; $2x - 4y = -24$
39. Solve the following pair of linear equations by elimination method :
 $x + 5y = 11$; $2x - 3y = 9$
40. If the sum of first 14 terms of an AP is 1050 and its first term is 10, find the 20th term.
41. Show that the points (1,7), (4,2), (-1, -1) and (-4,4) are the vertices of a square.
42. Find the ratio in which the y-axis divides the line segment joining the points (5, -6) and (-1, -4). Also find the point of intersection.
43. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
44. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
45. The following data gives the distribution of total monthly household expenditure of 200 families of a village :

Expenditure (in Rs)	Number of families
1000-1500	24
1500-2000	40
2000-2500	33
2500-3000	28
3000-3500	30
3500-4000	22
4000-4500	16
4500-5000	7

Find the modal monthly expenditure of the families.

46. A die is thrown twice. What is the probability that :
- (i) 5 will not come up either time?
 - (ii) 5 will come up at least once?

SECTION-D

Answer any 5 out of the following 8 questions with calculations.

(Question 47 to 54 : 4 Marks each)

Total : 20 Marks

47. State Basic Proportionality Theorem and prove it.
48. A girl of height 90 cm is walking away from the base of a lamp-post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, find the length of her shadow after 4 seconds.
49. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm., find the other two sides.
50. A manufacturer of TV sets produced 600 sets in the third year and 700 sets in the seventh year. Assuming that the production increases uniformly by a fixed number every year, find :
- the production in the 1st year.
 - the production in the 10th year.
 - the total production in first 7 year.

51. The table below shows the daily expenditure on food 25 households in a locality :

Expenditure (In Rs)	100-150	150-200	200-250	250-300	300-350
Number of households	4	5	12	2	2

Find the mean daily expenditure on food by a suitable method.

52. If the median of the distribution given below is 28.5. Find the values of x and y :

Class Interval	0-10	10-20	20-30	30-40	40-50	50-60	Total
Frequency	5	x	20	15	y	5	60

53. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting :
- A red face card
 - The jack of hearts
 - An ace of black colour
 - Not an ace

54. A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 (see the figure) and these are equally likely outcomes. what is the probability that it will point at :

(i) 8? (ii) an odd number? (iii) a number greater than 2? (iv) a number smaller than 9?



MATHEMATICS (BASIC)

Time: 3 Hours

Standard: 10th (EM)

Total Mark: 80

Date: 27/01/25

Instructions:

- (1) There are total 54 questions in this question paper and divided into four sections A, B, C and D.
- (2) General options are given in sections B, C and D.
- (3) Together with the instructions of each section, total marks of the corresponding section are given on the right side.
- (4) Start a new section on a new page. Answer the questions in serial order.
- (5) Draw figure, wherever necessary.
- (6) Use of calculator, smart watch or digital watch is strictly prohibited.
- (7) Write in neat and clean hand-writing.

SECTION-A

Answer the following as per the instruction given:

[24]

[Question no. 1 to 24, 1 mark each]

●Select the most appropriate answer from the given alternatives: (Question no. 1 to 6)

- 1) $\sqrt{9} + 3$ is a/an number
(a) rational (b) irrational (c) negative (d) fraction
- 2) HCF (18, 63) =
(a) 3 (b) 9 (c) 126 (d) 1
- 3) Number of zeros of $P(x) = x^3 - x$ is
(a) 0 (b) 1 (c) 2 (d) 3
- 4) If roots of the equation $ax^2 + bx + c = 0$ are inverse of each other, then
(a) $c = a$ (b) $c = -a$ (c) $a = -b$ (d) $a = b$
- 5) Two lines $x + 3y + 7 = 0$ and $2x + ky + 18 = 0$ do not intersect each other, then the value of k is
(a) 3 (b) 6 (c) 1 (d) 7
- 6) For $3x + 2y = 7$ and $2x + 3y = 3$, $x - y =$
(a) 10 (b) 4 (c) -4 (d) 21

- Fill in the blanks by selecting the proper answer from those given in the brackets to

Make the statements true: (Question no. 7 to 12)

- 7) is a solution of quadratic equation $x^2 - 10x - 24 = 0$. (4, 6, 12)
 8) For an A.P., if $a = 2$ and $d = 4$, then $a_{20} = \dots$ (600, 800, 78)
 9) $1 + 2 + 3 + \dots + 100 = \dots$ (5050, 10100, 20010)
 10) If P (0, 0) and P (-8, 6) then coordinates of its mid-point are ... [(-4,-3) (-4,3), (0,5)]
 11) $\cos^2 60^\circ = \dots\dots\dots$ ($\frac{1}{4}$, $\frac{3}{4}$, $\frac{1}{2}$)
 12) Two non-intersecting have common tangent (2, 3, 4)

- State whether the following statements are true or false: (Question no. 13 to 16)

- 13) $\cos (A+B) = \cos A + \cos B$
 14) The length of the segment of the tangent from the external point P and the point of contact with the circle is called the length of the tangent from the point P to the circle.
 15) If $P(A) = 0.01$, then $P(\text{not } A) = 0.90$
 16) If mean = median = 32, then mode = 16.

- Answer the following questions by a number or a word or a statement:

(Question no. 17 to 20)

- 17) Find the median of the observation : 5, 7, 6, 9, 8, 17, 4, 1.
 18) For the following information.

Class	0-10	10-20	20-30	30-40	40-50
Frequency	7	15	13	17	10

State the median class.

- 19) What is the probability of getting 5 Sunday in the month of October?
 20) What is the probability of getting less than 5 marks, in 100 marks paper?

- Match the following pairs correctly: (Question no. 21 to 24)

A	B
21. Volume of cone	(a) $\frac{1}{3} \pi r^2 h$
22. Volume of cylinder	(b) $2\pi r^2 h$
	(c) $\pi r^2 h$

A	B
23. Length of minor arc	(a) $2\pi r$
24. Circumference of a circle	(b) $\frac{\pi r^2 \theta}{360}$
	(c) $\frac{\pi r \theta}{180}$

SECTION-B

Solve any 9 questions from the following 13 questions:

[18]

[Question no. 25 to 37, 2 marks each]

- 25) 2 cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.
- 26) A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π .
- 27) Find the zeroes of the following quadratic polynomial $x^2 - 2x - 8$
- 28) Find a quadratic polynomial, the sum and product of whose zeroes are -3 and 2 , respectively.
- 29) Marks obtained out of 50, by 100 students in a test are given below. Find the median of the data:

Marks obtained	20	29	28	33	42	38	43	25
Number of students	6	28	24	15	2	4	1	20

- 30) Find the nature of the roots of the following quadratic equations. If the real roots exist, find them: $2x^2 - 3x + 5 = 0$.
- 31) The length of a tangent from a point A at distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.
- 32) Find the 10th term of the AP : 2, 7, 12, ...
- 33) Find the sum of the first 22 terms of the AP : 8, 3, -2 , ...
- 34) Find the distance between the points (2, 3) and (4, 1)
- 35) If (1, 2), (4, y), (x , 6) and (3, 5) are the vertices of a parallelogram taken in order, find x and y .
- 36) If tangents PA and PB from a point P to a circle with centre O are inclined to each other at angle of 80° , then find $\angle POA$
- 37) A piggy bank contains hundred 50p coins, fifty Re 1 coins, twenty Rs 2 coins and ten Rs 5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin (i) will be a 50 p coin ? (ii) will not be a Rs 5 coin?
-

SECTION-C

Solve any 6 questions from the following 9 questions as asked with calculations:

[Question no. 38 to 46, 3 marks each]

[18]

- 38) Prove that the lengths of tangents drawn from an external point to a circle are equal.
39) Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
40) A survey was conducted by a group of students as a part of their environment awareness programme, in which they collected the following data regarding the number of plants in 20 houses in a locality. Find the mean number of plants per house.

Number of plants	0-2	2-4	4-6	6-8	8-10	10-12	12-14
Number of houses	1	2	1	5	6	2	3

- 41) A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarized it in the table given below. Find the mode of the data:

Number of cars	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
Frequency	7	14	13	12	20	11	15	8

- 42) The following distribution shows the daily pocket allowance of children of a locality. The mean pocket allowance is Rs 18. Find the missing frequency f .

Daily Pocket allowance(in Rs)	11-13	13-15	15-17	17-19	19-21	21-23	23-35
Number of children	7	6	9	13	f	5	4

- 43) One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting
(i) a king of red colour (ii) a face card (iii) a red face card
(iv) the jack of hearts (v) a spade (vi) the queen of diamond
44) A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears
(i) a two-digit number (ii) a perfect square number (iii) a number divisible by 5.
45) An unbiased coin is tossed thrice. Find the probabilities of the following events:
(i) A: exactly two heads are received
(ii) B: at least two heads are received
(iii) C: at most two heads are received
46) Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

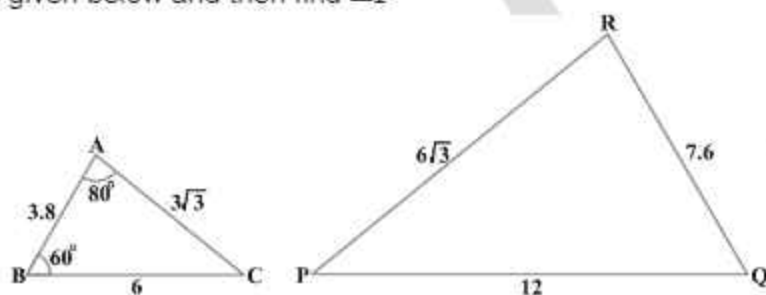
SECTION-D

Solve any 5 questions from the following 8 questions as asked with calculations:

[Question no. 47 to 54, 3 marks each]

[20]

- 47) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.
- 48) A girl of height 90 cm is walking away from the base of a lamp-post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, find the length of her shadow after 4 seconds.
- 49) If a line divides any two sides of a triangle in the same ratio, then prove that the line is parallel to the third side.
- 50) Observe the figure given below and then find $\angle P$



- 51) The median of the following data is 525. Find the values of x and y , if the total frequency is 100.

Class intervals	Frequency
0 - 100	2
100 - 200	5
200 - 300	x
300 - 400	12
400 - 500	17
500 - 600	20
600 - 700	y
700 - 800	9
800 - 900	7
900 - 1000	4

- 52) A life insurance agent found the following data for distribution of ages of 100 policy holders. Calculate the median age, if policies are given only to persons having age 18 years onwards but less than 60 year.

Age(in years)	Number of policy holders
Below 20	2
Below 25	6
Below 30	24
Below 35	45
Below 40	78
Below 45	89
Below 50	92
Below 55	98
Below 60	100

- 53) Two unbiased dice are thrown once. Write the possible outcomes of this experiment. What is the probability that the sum of the two numbers on two dice is :
- (a) 8
 - (b) Greater than 9
 - (c) Smaller than 5
-

MATHEMATICS (BASIC)

Time: 3 Hours

Standard: 10th (EM)

Total Mark: 80

WST-3

Date: 03/02/25

SECTION-A

Answer the following as per the instruction given:

[24]

[Question no. 1 to 24, 1 mark each]

● Select the most appropriate answer from the given alternatives: (Question no. 1 to 6)

1. If the HCF of two integers is 1, then the integers must be

- (A) prime (B) co-prime (C) composite (D) None of those

2. HCF of 96 and 404 = ?

- (A) 9096 (B) 9696 (C) 4 (D) 9006

3. The zeroes of the quadratic polynomial $x^2 + 7x + 12$ are

- (a) (-2, -5) (b) (-3, -4) (c) (2, 5) (d) (3, 4)

4. How many solutions do two linear equation in two variables have, if their lines in graph intersect in one point?

- (a) One solution (b) Two solutions
(c) Three solution (d) Infinite number of solution

5. Which of the following is a solution of $x + y = 5$, $2x + y = 3$?

- (a) $x = -2$, $y = 7$ (b) $x = 2$, $y = 3$ (c) $x = 1$, $y = 4$ (d) $x = 0$, $y = 5$

6. The nature of the roots of the quadratic equation $x^2 - 5x - 7 = 0$ is

- (a) imaginary (b) real and equal
(c) real and unequal (d) none of these

● Fill in the blanks by selecting the proper answer from those given in the brackets to Make the statements true: (Question no. 7 to 12)

7. $3 + \sqrt{16}$ is a number. (rational, irrational, negative integer)

8. The zero of polynomial $P(x) = \sqrt{5}x - 5$ is (5 , $\frac{1}{\sqrt{5}}$, $\sqrt{5}$)

9. Probability of the event that two dice are tossed simultaneously and the two number obtained on them are different $\left(\frac{5}{6}, \frac{1}{6}, \frac{2}{3}\right)$
10. As the value of θ increases, the value of $\sin \theta$ (increase, decrease, become negative)
11. Line intersect a circle in two points is called a (tangent, secant, parallel)
12. If the median is 30 and mean is 20 then the mode of the given data is (30, 40, 50)

• **State whether the following statements are true or false: (Question no. 13 to 16)**

13. The radii of two circles with centres O and O' are 9 cm and 7 cm and the distance between their centres is 15 cm then those circles will have four common tangents
14. A rectangle circumscribing a circle is a square
15. The formula to find the area of the quadrant of a circle is $\frac{1}{2} \pi r^2$
16. If point P lies in the exterior of the circle with centre O and radius r then $OP > r$
17. If the height and radius of a cone of volume V is doubled, then the volume of the cone becomes 8V.
18. If the radius of the base of a right circular cone is 3r and its height is equal to the radius of the base, then its volume is $9\pi r^3$.

• **Answer the following questions by a number or a word or a statement: (Question no. 17 to 20)**

19. If the mean of 3, 8, 7, a, 4, 9 is 6 then what is a?
20. Observations in ascending order are 24, 25, 26, x+2, x+3, 30, 31, 34 and its median is 27.5 then find x
21. If $M = 15$, $\bar{x} = 12$ then find z
22. Find the probability of getting 35 marks in a test of 50 marks
23. If $P(A) = 0.85$, then find $P(\bar{A})$
24. What is the probability of selecting at random a prime number from 1 to 100

SECTION-B

Solve any 9 questions from the following 13 questions:

[18]

[Question no. 25 to 37, 2 marks each]

25. Find a quadratic polynomial with the given numbers as the sum and product of its zeroes respectively: $\frac{1}{4}$, -1
26. Which term of the AP: 3, 8, 13, 18, is 78?
27. Find the 21st term of the AP: 3, 8, 13,

28. Find the zeros of $4x^2 - 4x + 1$

29. Find the roots of the following quadratic equations, if they exist, using the quadratic formula: $2x^2 - 6x + 3 = 0$

30. Find the distance between the pair of points: (7,5), (4,1)

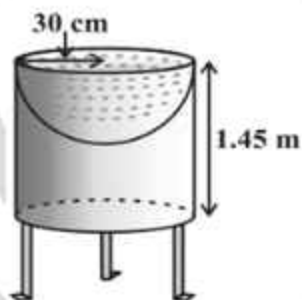
31. Check whether (5,-2), (6,4) and (7,-2) are the vertices of an isosceles triangle.

32. If $\sin A = \frac{3}{4}$, calculate $\cos A$ and $\tan A$.

33. Evaluate: $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$

34. The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30° . Find the height of the tower.

35. Mayank made a bird-bath for his garden in the shape of a cylinder with a hemispherical depression at one end (see Fig. 13.9). The height of the cylinder is 1.45 m and its radius is 30 cm. Find the total surface area of the bird-bath. (Take $\pi = \frac{22}{7}$)



36. A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy.

37. The marks obtained by 100 students of two classes in Mathematics paper consisting of 100 marks are as follows. Find median of the data

Marks obtained	15	20	25	32	35	45	50	60	70	77	80
No of students	2	3	7	4	10	12	9	8	6	8	11

SECTION-C

Solve any 6 questions from the following 9 questions as asked with calculations:

[Question no. 38 to 46, 3 marks each]

[18]

38. Solve the given pair of equations by elimination method.

$$2x + 3y = 46$$

$$3x + 5y = 74$$

39. The difference between two numbers is 26 and one number is three times the other. Find them.

40. In an AP, $a = 5$, $d = 3$, $a_n = 50$ Find ' n '.

41. Find the coordinates of the point which divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$.

42. Name the type of quadrilateral formed if any, by the following points and give reason for your answer

$$(-1, -2), (1, 0), (-1, 2), (-3, 0).$$

43. Prove that "the lengths of the tangents drawn from an external point to a circle are equal"

44. Two concentric circles are of radii 17 cm and 8 cm . Find the length of the chord of the larger circle which touches the smaller circle.

45. If a person holds the following shares in different companies, find their mode.

Shares	100- 200	200- 300	300- 400	400- 500	500- 600	600- 700
Companies	5	3	3	6	2	1

46. Two balanced dice are tossed once. Write the total number of possible outcome of this experiment and find the probability of the following events

(i) The sum of integers found on both dice is greater than 10

(ii) The numbers obtained on both dice are same..

SECTION-D

Solve any 5 questions from the following 8 questions as asked with calculations:

[Question no. 47 to 54, 3 marks each]

[20]

47. Prove that “if line is drawn parallel to the one side of a triangle to intersect the other two sides in distinct point, the other two sides are divided in the same ratio.”

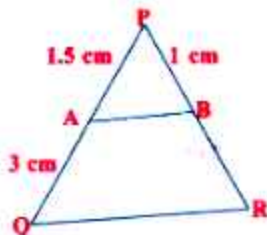
48. In ΔPQR , $QR \parallel AB$. Answer the following questions.

(i) Find BR .

(ii) Find PQ .

(iii) Find PR .

(iv) Which triangle is similar to ΔPAB ?



49. Find two consecutive positive integers, sum of whose squares is 365.

50. A manufacturer of TV sets produced 600 sets in the third year and 700 sets in the seventh year. Assuming that the production increases uniformly by a fixed number every year, find : (i) the production in the 1st year (ii) the production in the 10th year (iii) the total production in first 7 years

51. The table below shows the daily expenditure on food of 25 households in a locality.

Daily expenditure (in Rs.)	100-150	150-200	200-250	250-300	300-350
No. of households	4	5	12	2	2

Find the mean daily expenditure of food by a suitable method.

52. The given frequency distribution contains the power consumption information of 68 customers. Find the median of the data:

Monthly consumption (in unit)	65-85	85-105	105-125	125-145	145-165	165-185	185-205
No. of customers	04	05	13	20	14	08	04

53. One card is drawn from well shuffled deck of 52 card find the probability of getting:

(i) A face card (ii) Jack of heart (iii) A spade and (iv) queen of diamond.

54. A box contains 100 discs numbered from 1 to 100. If one disc is drawn from the box.

Find the probability that it bears

(i) A two digit number

(ii) A perfect square number

(iii) A number divisible by 5

(iv) A single digit number

Date: - 18/01/2025
Std: - 10th Eng. Med.

MATHEMATICS (STANDARD)

Time: 3 Hours
Total Mark: - 80

Instructions:

1. In all, **54** questions in this question paper are divided into **four** sections A, B, C and D.
2. The numbers on the right side represents the marks of that section.
3. Draw figure, wherever necessary. Maintain the lines and arcs of the construction.
4. Start writing a new section from a new page. Answer the questions in serial order.
5. Use of calculator is strictly prohibited.

SECTION-A

(Question 1 to 24 : 1 Mark each)

Total : 24 Marks

➤ Choose the correct option in the following :

1. The greatest integer that divides 70 and 125 leaving remainders 5 and 8 respectively is
(A) 13 (B) 65 (C) 875 (D) 1750
2. If α and β are the zeroes of polynomial $p(x) = 12x^2 + 19x + 5$, then $\frac{1}{\alpha} + \frac{1}{\beta} = \dots\dots$
(A) $-\frac{19}{12}$ (B) $\frac{5}{12}$ (C) $-\frac{5}{19}$ (D) $-\frac{19}{5}$
3. If $m + n = 14$ and $m - n = 4$, then $m = \dots\dots\dots$
(A) 18 (B) 10 (C) 9 (D) 56
4. If the discriminant of the equation $3x^2 - 4x - k = 0$ is 112, then $k = \dots\dots\dots$
(A) -8 (B) 8 (C) 32 (D) -32
5. The 4th term from the last term of the finite AP : - 11, - 8, - 5, 49, is
(A) 37 (B) 40 (C) 43 (D) 58
6. In trapezium ABCD, $AB \parallel CD$ and AC and BD intersect at point M. If $MA = 6$, $MB = 9$ and $MC = 8$, then $MD = \dots\dots\dots$
(A) $\frac{48}{9}$ (B) 12 (C) $\frac{54}{8}$ (D) 11

➤ **Fill in the blanks to make the statements correct.**

7. If the distance between the points $(2, -3)$ and $(5, b)$ is 5, then $b = \dots\dots\dots$
(2 or 5, 1 or -7 , -1 or 7)
8. If $7 \sin^2 \theta + 3 \cos^2 \theta = 3$, then $\sin \theta = \dots\dots\dots \left(\frac{1}{4}, 0, \frac{3}{4}\right)$
9. The sides of ΔABC measure 11 cm, 60 cm and 61 cm. Hence, the in radius of ΔABC is $\dots\dots\dots$ cm. (10, 5, 7)
10. The area of a quadrant of a circle with radius 21 cm is $\dots\dots\dots$ cm^2 (1386, 693, 346.5)
11. The radius of a cylinder is 21 cm and its curved surface area is 3960 cm^2 . Then, the height of the cylinder is $\dots\dots\dots$ (60, 30, 45)
12. If the mode of the observations 64, 60, 48, x , 43, 48, 43 and 34 is 43, then $x + 3 = \dots\dots\dots$
(43, 46, 48)

➤ **State whether the following are true or false?**

13. If $P(A) : P(\bar{A}) = 3 : 2$, then $P(A) = \frac{3}{5}$.
14. If $p\left(\frac{-2}{3}\right) = 0$, then $(3x + 2)$ is a factor of $p(x)$.
15. The roots of the equation $x^2 + 8x + 12 = 0$ are 6 and 2.
16. The vertices of a triangle are $(7, 6)$, $(7, 3)$ and $(3, 3)$. Then, the coordinates of the centroid of the triangle are $\left(\frac{17}{4}, 4\right)$

➤ **Answer the following in one word, sentence, number.**

17. What is the angle made by the hour hand in the clock in one minute?
18. If the mean of first n natural numbers is $\frac{17n}{33}$, find n .
19. State the smallest number which is divisible by 20 and 45 both.
20. If 3 is one of the roots of the equation $x^2 - kx + 36 = 0$, find the other root.

➤ **Match the following pairs correctly :**

'A'	'B'
21. $\alpha\beta\gamma$	(a) $-\frac{b}{a}$
22. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$	(b) $-\frac{d}{a}$
	(c) $-\frac{c}{d}$

'A'	'B'
23. $\tan \theta = \dots\dots\dots$	(a) $\frac{1}{\cos \theta}$
24. $\sec \theta = \dots\dots\dots$	(b) $\frac{\sin \theta}{\cos \theta}$
	(c) $\frac{\cos \theta}{\sin \theta}$

SECTION-B

Answer any 9 out of the following 13 questions with calculations.

(Question 25 to 37 : 2 Marks each)

Total : 18 Marks

25. Find the HCF and LCM of 510 and 92 and also verify that $\text{HCF} \times \text{LCM} = \text{the product of two given numbers}$.

26. Solve the following pair of linear equations by substitution method :

$$\frac{3x}{2} - \frac{5y}{3} = -2; \frac{x}{3} + \frac{y}{2} = \frac{13}{6}$$

27. Find the roots of the following quadratic equation by factorization : $2x^2 - x + \frac{1}{8} = 0$

28. Find the discriminant of the quadratic equation $3x^2 - 4\sqrt{3}x + 4 = 0$ and hence find the nature of roots. Find them, if they are real.

29. How many three digit numbers are divisible by 7?

30. Evaluate : $\frac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$

31. Prove that : $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$

32. Point A lies in the exterior of a circle with centre P and radius 7 cm. A tangent through A touches the circle at B. If $AB = 24$ cm. Find PA.

33. The sum of the radius of the base and height of a solid right circular cylinder is 37 cm. If its total surface area is 1628 cm^2 , find its volume.

34. For a given frequency distribution in usual notations. $l = 200$; $f_1 = 37$; $f_0 = 21$; $f_2 = 13$ and $h = 100$. Find the mode of the data.

35. A survey was conducted by a group of students as a part of their environment awareness programme, in which they collected the following data regarding the number of plants in 20 houses in a locality :

Number of plants	0 - 2	2 - 4	4 - 6	6 - 8	8 - 10	10 - 12	12 - 14
Number of houses	1	2	1	5	6	2	3

Find the mean number of plants per house.

36. In a carton of 20 bulbs, 4 bulbs are defective. Find the probability that a bulb selected at random is non-defective.

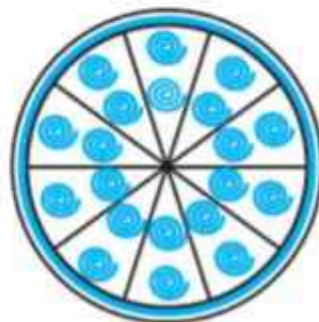
37. It is given that in a group of 3 students, the probability of 2 students not having the same birthday is 0.992. What is the probability that the 2 students have the same birthday?

SECTION-C

Answer any 6 out of the following 9 questions with calculations.
(Question 38 to 46 : 3 Marks each)

Total : 18 Marks

38. Find the zeroes of the polynomial $2x^2 + 11x + 15$ and verify the relationship between the zeroes and the coefficients.
39. Without finding the zeroes α and β of polynomial $p(x) = 6x^2 - 13x + 6$, find the values of the following :
- (i) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (ii) $\alpha^2 + \beta^2$ (iii) $\alpha^3 + \beta^3$
40. The 10th term of an AP is 52 and its 17th term is 20 more than its 13th term. Find the AP and its 30th term.
41. If the sum of first 7 terms of an AP is 49 and that of 17 terms is 289, find the sum of first n terms.
42. Find the ratio in which the point $(-3, p)$ divides the line segment joining the points $(-5, -4)$ and $(-2, 3)$. Also find the value of p .
43. If a, b, c are the sides of a right triangle, where c is the hypotenuse. Prove that the radius of the circle which touches all the sides of the triangle is given by $r = \frac{a + b - c}{2}$.
44. AB is tangent to a circle with centre P and B is the point of contact. PA intersects the circle at C. If AB = 15 cm and AC = 9 cm, find the radius of the circle.
45. A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors as shown in the given figure.
- Find :
- (i) the total length of the silver wire required.
- (ii) the area of each sector of the brooch.



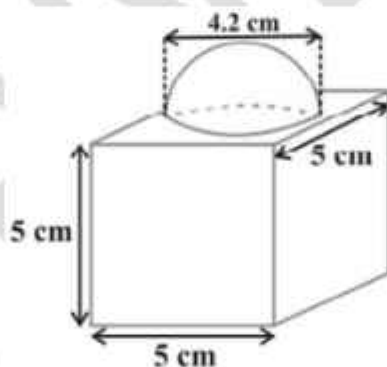
46. A jar contains 24 marbles, some are green and others are blue. If a marble is drawn at random from the jar. The probability that it is green is $\frac{2}{3}$. Find the number of blue marbles in the jar.

SECTION-D

Answer any 5 out of the following 8 questions with calculations.
(Question 47 to 54 : 4 Marks each)

Total : 20 Marks

47. The sum of a two digit number and the number obtained by reversing the digits is 66. If the digits of the number differ by 2, find the number. How many such numbers are there?
48. Prove converse Thales Theorem.
49. Prove Thales Theorem.
50. E is a point on the side AD produced of a parallelogram ABCD and BE intersects CD at F. Show that $\triangle ABE \sim \triangle CFB$.
51. As observed from the top of a 75 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45° . If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.
52. The decorative block shown in the given figure is made of two solids – a cube and a hemisphere. The base of the block is a cube with edge 5 cm, and the hemisphere fixed on the top has a diameter of 4.2 cm. Find the total surface area of the blocks.
(Take $\pi = \frac{22}{7}$)



53. A rocket is in the form of a right circular cylinder closed at the lower end and surmounted by a cone of the same radius. The diameter and height of the cylinder are 6 cm and 12 cm respectively, while slant height of the cone is 5 cm. find the volume of the rocket. (Use $\pi = 3.14$)
54. To find out the concentrations of SO_2 in the air (in parts per million, i.e. ppm) the data was collected for 30 localities in a certain city and is presented below:

Concentration of SO_2 (in ppm)	0.00 – 0.04	0.04 – 0.08	0.08 – 0.12	0.12 – 0.16	0.16 – 0.20	0.20 – 0.24
Frequency	4	9	9	2	4	2

Find the mean concentration of SO_2 in the air.

Date: - 27/01/2025
Std: - 10th Eng. Med.

MATHEMATICS (STANDARD)

Time: 3 Hours
Total Mark: - 80

Instructions:

1. In all, **54 questions** in this question paper are divided into **four sections A, B, C and D**.
2. The numbers on the right side represents the marks of that section.
3. Draw figure, wherever necessary. Maintain the lines and arcs of the construction.
4. Start writing a new section from a new page. Answer the questions in serial order.
5. Use of calculator is strictly prohibited.

SECTION-A

(Question 1 to 24 : 1 Mark each)

Total : 24 Marks

➤ Choose the correct option in the following :

1. A pair of linear equations which has a unique solution $x = 2, y = -3$ is...
(A) $x + y = -1, 12x - 3y = -5$ (B) $2x + 15y = 5, 4x + 10y = -22$
(C) $2x - y = 17, 3x + 2y = 12$ (D) $x - 4y - 14 = 0, 5x - y - 13 = 0$
2. The probability of the month February of a non-leap year having 5 Fridays is
(A) $\frac{3}{7}$ (B) $\frac{2}{7}$ (C) $\frac{1}{7}$ (D) 0
3. The centroid of the triangle with vertices (4, 8), (-9, 7) and (8, -3) is
(A) (4, 1) (B) (1, 4) (C) (-5, 4) (D) (5, 7)
4. 8, a, 16 are in AP, then $a =$
(A) 8 (B) 10 (C) 12 (D) 14
5. The ratio of radii of two cones with equal height is 2 : 3 then, the ratio of their volumes is
(A) 4 : 6 (B) 8 : 27 (C) 3 : 2 (D) 4 : 9
6. The cubic polynomial $p(x) = 27x^3 + 1$ has real zeros.
(A) 0 (B) 1 (C) 2 (D) 3

➤ **State whether the following are true or false?**

7. The general difference of an AP 2.5, 4, 5.5, is a positive integer.
8. TSA of cone is $\pi r(r + l)$ square units.
9. Walking 40m on a slope of inclination 30° with the level ground, one reaches 10 m above the ground.
10. For any frequency distribution, $\text{mode} - 3\text{median} = 2\text{mean}$.
11. Graph of $7x - 3y = 4$ passes through (1, 1).
12. If a line PA intersects the circle with centre O, at two distinct points, then PA is tangent.

➤ **Fill in the blanks to make the statements correct.**

13. The tangent drawn from point P to a circle with centre O is PQ. If $\triangle OPQ$ is an isosceles triangle, then $\angle OPQ = \underline{\hspace{2cm}}$. ($45^\circ, 60^\circ, 90^\circ$)
14. In $P(x) = mx^2 - 5x + 3$ sum of zeroes is 2.5, then $m = \underline{\hspace{2cm}}$. (1, 2, 3)
15. One of the zero of $x^2 - kx + 12 = 0$ is 6, then other zero is $= \underline{\hspace{2cm}}$. (12, 8, 2)
16. In $\triangle XYZ$, $\angle Y = 90^\circ$, $XZ = 13$ cm and $YZ = 5$ cm, then inradius of $\triangle XYZ = \underline{\hspace{2cm}}$ cm. (5, 2, 3)
17. H.C.F (56, 112) = $\underline{\hspace{2cm}}$. (56, 1, 112)
18. $\sum_{i=1}^{10} x_i - \bar{x} = \underline{\hspace{2cm}}$. (0, 1, $9\bar{x}$)

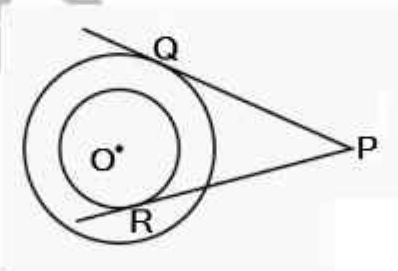
➤ **Answer the following in one word, sentence, number.**

19. In $\triangle ABC$, $\angle ABC = 90^\circ$, $\angle ACB = 45^\circ$ and $AC = 6$, then find area of $\triangle ABC$.
20. If $\sin \theta = \frac{1}{2}$, then what is the value of $\frac{\cos \theta}{\sqrt{3}} + \sin \theta$?
21. Write the TSA of one-fourth of a lemon of radius r.
22. Find the probability of getting 35 marks in a test of 50 marks.
23. Write the sum of first 10 multiples of 7.
24. If a girl of height 1.5 m stands at a distance of 3 m from a lamp post and casts a shadow of 4.5 m on the ground, then what is the height of the lamp post?

SECTION-B

Answer any 9 out of the following 13 questions with calculations.
(Question 25 to 37 : 2 Marks each)

Total : 18 Marks

25. Using H.C.F $(a, b) \times \text{L.C.M}(a, b) = ab$, find L.C.M $(64, 28)$.
26. Prove that $7 - 3\sqrt{2}$ is irrational.
27. Find the roots of the quadratic equation $3x^2 - 3\sqrt{6}x + 2 = 0$ by factorization.
28. How many three digit number are divisible by 7 ?
29. Which term of A.P.: 8,11,14,17.....is 272 ?
30. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$ $0^\circ < A + B \leq 90^\circ$, $A > B$. Find A and B.
31. Radii of two concentric circles are 13 and 12. A Chord of the circle with larger radius Touches the circle with smaller radius. Find the length of the chord.
32. Find the area of a sector of a circle with radius 6 cm if angle of the sector is 60° .
33. The radius of two concentric circles in the given figure is 5 cm and 3 cm, PQ and PR are tangents from the points Q and R of those circles. If $PQ = 12$ cm, find PR.
- 
34. 2 cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.
35. For a given frequency distribution, in usual notations, $n = 60$, $cf = 10$, $h = 5$ & $\ell = 40$. Find the median of the data.
36. Two balanced dice are rolled. Find the probability that the sum of numbers on two dice is a multiple of 7.
37. A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . Find the height of the tower.

SECTION-C

Answer any 6 out of the following 9 questions with calculations.

(Question 38 to 46 : 3 Marks each)

Total : 18 Marks

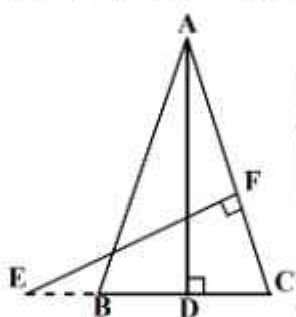
38. Find the zeroes of polynomial $12x^2 + 5x - 2$ and hence verify the relationship between zeroes and coefficients.
39. Solve the pair of linear equations $3x + 4y = -14$ and $x - 5y = 22$ by the method of Substitution.
40. Find two numbers whose sum is 27 and product is 182.
41. Find the coordinates of the point which divides the line segment joining the points A(4, -3) and B(8, 5) in the ratio 3:1 internally.
42. Find a point on the y-axis which is equidistant from the points A(6, 5) and B(-4, 3).
43. A chord of a circle of radius 10 cm subtends right angle at the centre. Find the area of the corresponding (i) minor segment (ii) major sector. ($\pi = 3.14$).
44. Evaluate : $\frac{\sin 30^\circ}{\cos 45^\circ} + \frac{\tan 45^\circ}{\sec 60^\circ} - \frac{\cos 30^\circ}{\tan 45^\circ} - \frac{\sin 60^\circ}{\sin 90^\circ}$.
45. One card is drawn from a well shuffled deck of 52 cards. Find the probability of getting
(i) A king of red colour (ii) A face card (iii) The jack of hearts
46. A tree breaks due to a storm & the broken part bends such that the top of the tree touches the ground making an angle having measure 30 with ground. The distance from the foot of the tree to the point where the top touches ground is 30m. Find the height of tree.

SECTION-D

Answer any 5 out of the following 8 questions with calculations.
(Question 47 to 54 : 4 Marks each)

Total : 18 Marks

47. The angles of depression of the top and the bottom of an 8m tall building from the top of a multi-storeyed building are 30° and 45° , respectively. Find the height of the multi-storeyed building and the distance between the two buildings.
48. The ten's digit of a two digit number is thrice the unit digit. The new number obtained by interchanging the digits is 54 less than the original number. Find the original number.
49. In figure, E is a point on side CB produced of an isosceles triangle ABC with $AB = AC$. If $AD \perp BC$ and $EF \perp AC$, prove that $\triangle ABD \sim \triangle ECF$.



50. State and prove Thales theorem.
51. A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1 m and 4 m respectively, and the slant height of the top is 2.8 m., find the area of the canvas used for making the tent. Also, find the cost of the canvas of the tent at the rate of Rs 500 per m^2 . (Note that the base of the tent will not be covered with canvas.)
52. A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm^3 of iron has approximately 8g mass. (Use $\pi = 3.14$).
53. Find the median of the following frequency distribution.

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency	2	3	3	4	3	4	7	11	8	10

54. The mode of the following frequency distribution of 165 observations is 34.5. Find the value of a & b.

Class	5-14	14-23	23-32	32-41	41-50	50-59	59-68
Frequency	5	11	a	53	b	16	10

Date: - 03/02/2025
Std: - 10th Eng. Med.

MATHEMATICS (STANDARD)

Time: 3 Hours
Total Mark: - 80

Instructions:

1. In all, **54 questions** in this question paper are divided into **four sections A, B, C and D**.
2. The numbers on the right side represents the marks of that section.
3. Draw figure, wherever necessary. Maintain the lines and arcs of the construction.
4. Start writing a new section from a new page. Answer the questions in serial order.
5. Use of calculator is strictly prohibited.

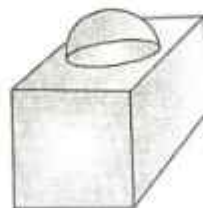
SECTION-A

(Question 1 to 24 : 1 Mark each)

Total : 24 Marks

➤ Choose the correct option in the following :

1. If p, q and r are prime numbers, $\text{LCM}(p, q, r) : \text{GCD}(p, q, r) = \dots\dots$
(A) $1 : pqr$ (B) $pqr : 1$ (C) 1 (D) None
2. As shown in the figure, one show- piece is made up of cube and hemisphere. If total surface area of a cube is denoted by A , curved surface area of hemisphere by B and area of the base of hemisphere by C , the total surface area of a showpiece is
(A) $A + B + C$
(B) $A + B - C$
(C) $B + C - A$
(D) $A + C - B$
3. For a frequency distribution $A = 31.6$, $\Sigma f_i u_i = 12$, $n = 24$ & $c = 12$, then $\bar{x}$ is.....
(A) 37.6 (B) 37.0 (C) 43.6 (D) None
4. gave the general formula to solve the quadratic equation.
(A) Euclid (B) Shidharacharya (C) Aryabhatta (D) Bhaskaracharya
5. The graph of is a line.
(A) $p(x) = x^2 + 8$ (B) $p(x) = x^3 - 2$
(C) $p(x) = x^2 + 6x + 9$ (D) $p(x) = 5x - 10$
6. M is the foot of perpendicular on x - axis from $P(-3, 2)$, then coordinates of M are....
(A) $(0, 2)$ (B) $(3, 0)$ (C) $(-3, 0)$ (D) None



➤ **State whether the following are true or false?**

7. If two circles touch each other externally, then they have two common tangents.
8. For a quadratic equation, if D is less than zero, then it has real zeroes.
9. In usual notations, if one zero of a quadratic equation is zero, then $c = 0$.
10. Two quadrilaterals are similar, if their corresponding sides are proportional.
11. If the graph of a polynomial intersects the x -axis at exactly two points, then it has two zeroes.
12. For an acute angle A , $\tan A$ is always less than 1.

➤ **Fill in the blanks to make the statements correct.**

13. The probability of getting a factor of 6 in a single throw of dice is _____. $\left(\frac{2}{3}, \frac{1}{2}, 0\right)$
14. If the median of the observations arranged in the ascending order as 6, 7, $x - 2$, x , 17, 20 is 16 then $x =$ _____. (17, 19, 21)
15. If $x = -2$ is a zero of $P(x) = 3x^2 + 12x - 2a$, then $a =$ _____. (6, -2, -6)
16. $\sum_{i=1}^n x_i - \bar{x} =$ _____. (0, $(n - 1)\bar{x}$, 1)
17. If $a_n = 9n - 21$, then $a_3 =$ _____. (6, 9, 30)
18. In triangle ABC, if $AB = 12$ cm, $BC = 16$ cm and $AC = 20$ cm, then its inradius is _____ cm. (4, 8, 1)

➤ **Answer the following in one word, sentence, number.**

19. If α, β, γ are the zeros of the cubic polynomial $p(x) = x^3 - 3x + 2x^2 + 5$, then find the value of $\alpha\beta + \beta\gamma + \gamma\alpha$.
20. If (4, 3) is one of the solutions of the equation $3x + 4y - 5k = 49$, then find the value of k .
21. Write the condition so that given pair of linear equation have consistent solution
 $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$
22. One letter is chosen randomly from the word PROBABILITY. Find the probability of the letter being a vowel.
23. The ratio of the diameter of two spheres is 4 : 3, then find the ratio of their volumes.
24. If $P(\bar{A}) : P(A) = 4 : 7$, then find the value of $P(A)$.

SECTION-B

Answer any 9 out of the following 13 questions with calculations.

(Question 25 to 37 : 2 Marks each)

Total : 18 Marks

25. Find the HCF of 96 and 404 by the prime factorisation method. Hence, find their LCM.
26. For what values of k will the following pair of linear equations have infinitely many solutions? $kx + 3y - (k - 3) = 0$ and $12x + ky - k = 0$
27. Find the zeroes of the polynomial $x^2 - 3$ and verify the relationship between the zeroes and the coefficients.
28. Find the discriminant of the equation $3x^2 - 2x + \frac{1}{3} = 0$ and hence find the nature of its roots. Find them, if they are real.
29. Check where 301 is a term of the list of numbers 5, 11, 17, 23,
30. Evaluate : $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ - \cos^2 30^\circ}$
31. Prove that : $\frac{\sin \theta}{1 - \cos \theta} = \operatorname{cosec} \theta + \cot \theta$
32. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
33. The cost of painting the surface of a sphere is Rs.1526 at the rate of Rs.6 per m^2 . Find the radius of the sphere.
34. For some given data, if $M - Z = 1.4$ and $M + Z = 75$, then find mean, median and mode.
35. The following data gives the information on the observed lifetimes (in hours) of 225 electrical components :

Life time (In hours)	0-20	20-40	40-60	60-80	80-100	100-120
Frequency	10	35	52	61	38	29

Determine the modal lifetimes of the components.

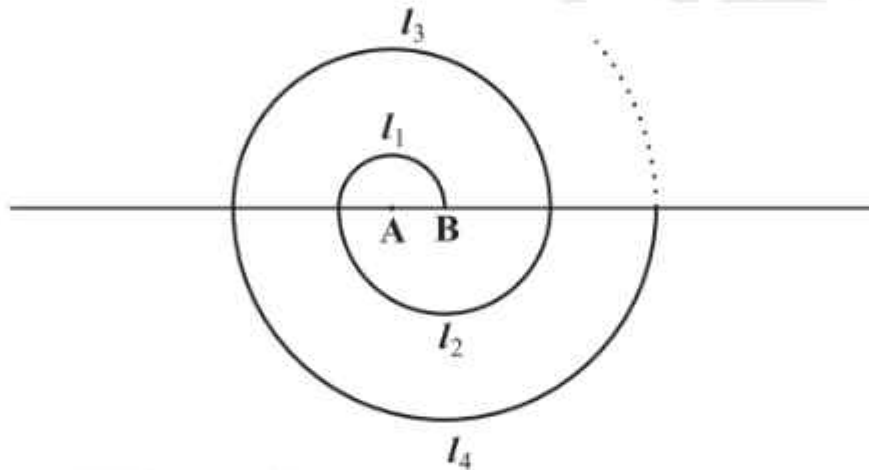
36. A die is thrown once. What is the probability of getting :
- (i) A prime number (ii) A number lying between 2 and 6
37. A box contains 90 discs numbered from 1 to 90. If one disc is drawn from the box. Find the probability that it bears:
- (i) A two digit number (ii) A perfect square number

SECTION-C

Answer any 6 out of the following 9 questions with calculations.
(Question 38 to 46 : 3 Marks each)

Total : 18 Marks

38. Find the zeroes of polynomial $6x^2 - 11x - 7 = 0$, hence verify the relationship between zeroes and coefficients.
39. The combined weight of father and son is 70 kg and the son's weight is one sixth of the father's weight. Find the weight of father and son.
40. A spiral is made up of successive semicircles, with centres alternately at A and B, starting with centre at A, of radii 2 cm, 4 cm, 6 cm, 8 cm, as shown in figure. What is the total length of such a spiral made up of thirteen consecutive semicircles? (Take $\pi = 22/7$)



41. In an AP, given $a = 8$, $a_n = 62$, $S_n = 210$, find n and d .
42. If $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$ are the vertices of a parallelogram taken in order, find x and y .
43. If r is the in radius of $\triangle ABC$ in which $\angle B = 90^\circ$, then prove that $r = \frac{AB + BC - AC}{2}$.
44. Prove that "The lengths of the tangents drawn from an external point to a circle are equal"
45. A chord of a circle of radius 21 cm subtends an angle of 120° at the centre. Find the area of corresponding minor segment of the circle. (Take $\pi = 22/7$)
46. One card is drawn from a well shuffled deck of 52 cards. Find the probability of getting :
- (i) A king of red colour
 - (ii) A face card
 - (iii) The queen of diamonds

SECTION-D

Answer any 5 out of the following 8 questions with calculations.
(Question 47 to 54 : 4 Marks each)

Total : 20 Marks

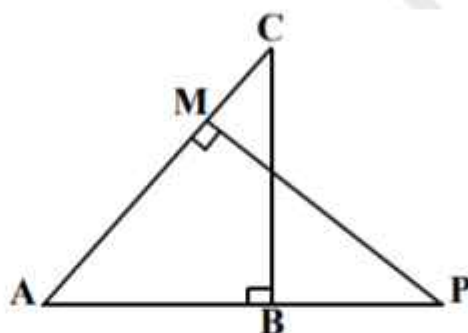
47. The sum of the digits of a two-digit number is 9. Also, nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

48. Rohan's mother is 26 year older than him. The product of their ages (in year) 3 years from now will be 360. Find their present ages.

49. In figure, ABC and AMP are two right triangles, right angled at B and M respectively. Prove that:

(i) $\triangle ABC \sim \triangle AMP$

(ii) $\frac{CA}{PA} = \frac{BC}{MP}$



50. State and prove converse of Thales theorem.

51. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the time taken by the car to reach the foot of the tower from this point.

52. A hemispherical depression is cut out from one face of a cubical wooden block such that the diameter of the hemisphere is equal to the edge of the cube. Determine the surface area of the remaining solid.

53. A gulab jamun, contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends with length 5 cm and diameter 2.8 cm (see figure).



54. The table below shows the daily expenditure on food of 25 households in a locality.

Daily expenditure (In Rs)	100-150	150-200	200-250	250-300	300-350
Number of households	4	5	12	2	2

Find the mean daily expenditure on food by a suitable method.

विषयः : संस्कृत

दिनाङ्कः २१/०१/२०२५

पाठ्यक्रमः 1 to 20 / व्याकरणम्

Standard- दशमीकक्षा

गुणाङ्कः : ८०

अभिधानम्:

कक्षा: क्रमाङ्कः-

Section-A – गद्यविभाग (16)

गद्यखण्डस्य अनुवादं कुरुत। (any one)

(4)

1. गवादीनामिव अचेतनेषु अपि पदार्थेषु नियमोऽयं दृश्यते। तद्यथा-लोष्टः क्षिप्तो बहुवेगं गत्वा नैव तिर्यग् गच्छति नोर्ध्वमारोहति। सः पृथिव्याः विकारः वर्तते इति कृत्वा पृथिवीमेव आगच्छति, सादृश्यत्वात्। तथा ज्योतिषो विकारः अर्चिः आकाशदेशे निवाते सुप्रज्वलितं नैव तिर्यग् गच्छति नार्वागारोहति। ज्योतिषो विकारो ज्योतिरेव गच्छत्यान्तर्यतः।

2 अथ तेषां तत्कुलशोपमं वचनं समाकर्ण्य मत्स्याः परस्परं प्राहुः - "श्रुतोऽयं धीवरालापः। अधुना अस्माभिः किं कर्तव्यम् ?" तत्र अनागतविधाता नाम मत्स्यः प्राह - "नूनं प्रभातसमये धीवराः अत्रागम्य मत्स्यसंक्षयं करिष्यन्ति इति मम मनसि वर्तते । तन्न युक्तं साम्प्रतं क्षणमपि अत्र अवस्थातुम् ।"

संस्कृतप्रश्नानाम् उत्तराणि विकल्पेभ्यःचित्वा लिखत। (any two)

(2)

3. श्रेष्ठीपुत्रः किदृशीं भार्यां इच्छति ? A.रूपवतीः B.गुणवतीः C.कलावतीः

4. प्रवाहरूपे परिवारे किं वहति ? A.स्नेहरूपं आदरम् B.स्नेहरूपं प्रवाहम् C.स्नेहरूपं जलम्

5. कुपितस्य स्वेदसलिलात् पूर्वं किं गलति? A.तपः B.रागः C.अधरः

प्रश्नानाम् उत्तराणि आङ्गलभाषायां लिखत।

(any two)

(4)

6.Which things does the girl sell ? what does she get in return?

7.Why did Brahma compare Durvasa with an actor? 8. Who does not deserve real happiness?

संक्षेपेण टिप्पणी लिखत। (2)

9.Obstacle named avartapaat- falling in whirlpool. 10. Frauds Of kutnatha.

गद्यखण्डं पठित्वा प्रश्नानाम् उत्तराणि लिखत। (4)

11.अथ तेषां तत्कुलिशोपमं वचनं समाकर्ण्य मत्स्याः परस्परं प्राहुः- “श्रुतोऽयं धीवरालापः। अधुना अस्माभिः किं कर्तव्यम्।” तत्र अनागतविधाता नाम मत्स्यः प्राह-“नूनं प्रभातसमये धीवराः अत्रागम्य मत्स्यसंक्षयं करिष्यन्ति इति मम मनसि वर्तते। तन्न युक्तं साम्प्रतं क्षणमपि अत्र अवस्थातुम्। अस्माभिः राज्ञी एव किञ्चित् समीपं सरः गन्तव्यम्। अहं तावत् जलाशयान्तरं गच्छामि।”

(1) धीवराणां वचनं कीदृशम् आसीत्? (2) कः मत्स्यः जलाशयान्तरं गन्तुम् इच्छामि?

(3) धीराः कदा आगमिष्यन्ति ? (4)नूनम् शब्दस्य समानार्थकं-_____ (खलु सदस्यः कदा)

Section-B – पद्यविभाग (16)

पद्यखण्डस्य अनुवादं कुरुत। (4)

12. शतं पदानि निष्क्रम्य शौचार्थं गच्छ सत्वरम्। फेनिलेन करीं कृत्वा शुद्धीं व्यायाममाचर ॥ १ ॥

तैलं मर्दय काये त्वं ततः स्नानं समाचर। यथाविधि जलेनैव वस्त्रं स्वीयं प्रधावय ॥ २ ॥

13. इत्युक्तः क्रोधतामाक्षः तप्तकाञ्चनकुण्डलतः। राक्षसेन्द्रोऽभिदुद्राव पतगेन्द्रम् अमर्षणः॥

तद् बभूवादभुतं युद्धं गृधराक्षसयोस्तदा। सपक्षयोर्माल्यवतोः महापर्वतयोरिव॥

संस्कृतप्रश्नानाम् उत्तराणि विकल्पेभ्यःचित्वा लिखत। any two (2)

14. कविः किं याचते? A.भद्रम् B.दुरितम् C.सुखम्

15. कः क्षीभरहितः भवतु ? A अयं ग्रामम्. B.अयं मण्डलः c.अयं देशः

16. कार्य प्रारम्भ के न परित्यजन्ति ? A. मध्यमजनाः, B. उत्तमजनाः C. नीचजनाः

प्रश्नानाम् उत्तराणि आङ्ग्लभाषायां लिखत| any three

(6)

17. In what way should a wife talk with her husband? 18. In how many ways is man tested?

19. Which adjective does a devotee use for himself in the stanza अनाथो दरिद्रो..?

20. How can anyone become बहुव्रीहिः?

श्लोकपूर्तिं कुरुत| any two

(4)

21. आपत्सु रामः भवार्जनेयः 22. अनाथो दरिद्रो.....त्वमेका भवानी| 23. यस्मान्नोद्विजते...मे प्रियः

Section-C – नाट्यविभाग (16)

नाट्यखण्डस्य अनुवादं कुरुत| any one

(4)

24. पुनीतः - द्वितीयं कार्यं तु कर्तव्यस्य निर्वाहः। दुर्घटनाग्रस्तः जनः सद्यः एव औषधालयस्थितः करणीयः भवति येन तस्य उचितः उपचारः स्यात्। तत्र सङ्गत्य कदाचित् तस्मै रक्तस्य आवश्यकता भवति चेत्, स्वकीयं रक्तमपि तस्मै दातव्यं भवति। तस्य परिवारे सूचना दातव्या भवति येन तस्य परिवारजनः औषधालयमागत्य तस्य विशेषः सहायो भवेत्। दुर्घटनायाः साक्षिणः मनुष्यस्य एकादृशं कर्तव्यम्। सर्वैः मनुष्यैः अस्य कर्तव्यस्य निर्वाहः करणीयः एव।

सुनीतः - तृतीयं कार्यं किमस्ति, यत् घटनायाः साखीं मनुष्यः आचरेत्?

25. शाण्डिल्यः - ओ भगवन् । इदमुद्यानम् ।

परिव्राजकः - प्रविश अग्रतः।

शाण्डिल्यः - भगवान् एव पुरतः प्रविशतु। अहं पृष्ठतः प्रविशामि।

परिव्राजकः - किमर्थम् ?

शाण्डिल्यः - पौराणिक्याः मम मातुः श्रुतम् अशोक- पल्लवान्तरनिरुद्धो व्याघ्रः प्रतिवसति । तुत् भगवानेव

पुरतः प्रविशतु अहं पृष्ठतः प्रविशामि।

वाक्यानि कः वदति इति लिखत। any two

(2)

"अ"

"ब"

26.न वयं द्रुतघातकाः

चाणक्यः

27.द्वितीयं कार्यं किमस्ति?

दुर्योधनः

28.उपाध्यायः अयं श्रेष्ठी चन्दनदासः

सुनीतः

शिष्यः

प्रश्नानाम् उत्तराणि आङ्ग्लभाषायां लिखत। any two

(4)

29. What does ghatotkacha advise shakuni to do?

30. Which man is compared with a stone? Why?

31. How does chanakya threaten chandandasa ?

संक्षेपेण टिप्पणीं लिखत। any two

(4)

32. Ghatotkacha

33. third duty of a man as a witness

34. character sketch of chandandasa

कोष्ठगतपदानि प्रयुज्य संस्कृत वाक्यानि रचयत । any two

(2)

35. I ghatotkacha bow down. (अस्मद् घटोत्कच अभि + वद)

36. An accident does not occur without reason. (कारण विना दुर्घटना न भू)

37. You are protecting the family of Rakshasa. (युष्मद् राक्षस गृहजन परिवार रक्ष)

Section-D – व्याकरणविभाग (16)

सुचनानुसारं उत्तराणि लिखत।

38.समानार्थ शब्दं लिखत। भार्या:-_____ (मातुलानी,स्वसा, जाया) (1)

39.विरुद्धार्थ शब्दं लिखत। विषाद:-_____ (शोकः, प्रसन्नताः, उपासना) (1)

40.वर्तमानकालस्य उत्तम पुरुष एकवचनम् रूपं लिखत। (मन्यावहे,मन्ये,मन्येते) (1)

41.भविष्यकालस्य अन्य पुरुष द्विवचनम् रूपं लिखत। (बोधिष्येते,बोधिष्यते,बोधिष्यामि) (1)

42.आनार्थकालस्य अन्य पुरुष बहुवचनम् रूपं लिखत।(विद्यताम् विद्येताम् विद्यन्ताम्) (1)

43.विध्यर्थकालस्य अन्य पुरुष एकवचनम् रूपं लिखत। (पिबेम,पिबेत्,पिबेः) (1)

44.स्म प्रयोगं कुरुत। "सः अचिन्तयत् " (1)

(क)सो चिन्तयति स्म। (ख) सः चिन्तयति स्म। (ग) सः चिन्तयामि स्म।

नामरूपेभ्यः योग्यं रूपं चित्वा लिखत। (2)

45.षष्ठी विभक्ति बहुवचनम्- _____ (रमाभ्यः, रामाणाम्, रमे)

46.द्वितीया विभक्ति एकवचनम् - _____ (पाठशालां , पाठशालायै, पाठशालाभिः)

47.उपपद विभक्तेः प्रयोगं कृत्वा रिक्तस्थानं पूरयत। (1)

_____नमः। (शिवं, शिवाय, शिवस्य)

48.विकल्पेभ्यः संख्यापदं चित्वा रिक्तस्थानं पूरयत। (1)

त्वं_____ पदानि निष्क्रिय शौचार्थं गच्छ ।(शतम् शतानि सप्तशतम्)

49.विकल्पेभ्यः कृदन्तप्रकारं चित्वा उत्तरं लिखत। (1)

तत् अधिकं शोचनीयम् अस्ति।(विध्यर्थ कर्मणि कृदन्त, कर्तरिभूतकृदन्त, सम्बन्धक कृदन्त)

50.सन्धि विच्छेदं कुरुत। (1)

देशोऽयं - _____ (देशाः+ अयम्,देशोः+अयम्,देशः+ अयम्)

51.सन्धिं योजयत। (1)

जलेनः+ एव- _____ (जलेनाव ,जलेनैव, जलेनाऽव)

सामासिक प्रकारं लिखत। (2)

52.जनसमूहम्:- _____ (पञ्चमी तत्पुरुष, तृतीया तत्पुरुष, षष्ठी तत्पुरुष)

53.संयतेन्द्रियः:- _____ (कर्मधारय, चतुर्थी तत्पुरुष, इतरेतर द्वन्द्व)

Section-E – संयुक्तविभाग (16)

54.वाक्यानि कथानक क्रमानुसारेण लिखत। (4)

(1) जगति यत्समुत्पन्नं नश्यति सर्वथा ध्रुवम्।

(2) अस्ति वित्तदासो नाम प्रभूतलोभः समृद्धः कश्चित् वणिग्जनः।

(3) एतानि पात्राणि मम न सन्ति

(4) भौ वञ्चक ! त्वं मिथ्या वदसि ।

55. गद्यखण्डं पठित्वा प्रश्नानाम् उत्तराणि लिखत। (4)

परोपकारशीलाः जनाः विश्वस्य कल्याणाय कार्यं कुर्वन्ति । ते परेषाम् उपकाराय घोरतरं कष्टं सहन्ते । यथा

दानशीलः कर्णः स्वकवचकुण्डले अयच्छत्। राजा शिबिः कपोतस्य प्राणान् रक्षितुं निजमांसम् अयच्छत्।

दधीचिः देवानां रक्षायै निजशरीरम् अपि अयच्छत्। ईदृशाः जनाः परेषां कृते उदाहरणस्वरूपाः सन्ति। तेषां कार्याणि अनुकरणीयानि ।

(1) कीदृशाः जनाः विश्वस्य कल्याणाय कार्यं कुर्वन्ति ? (2) कः देवानां रक्षायै निजशरीरम् अयच्छत् ?

(3) राजा शिविः किमर्थं स्वमांसम् अयच्छत् ? (4) दानशीलः कर्णः किम् अयच्छत् ?

कर्ता - कृतिनां योग्यं मेलनं कुरुत। (2)

“अ”

“ब”

56. महाकवि कालिदासः.

अष्टाध्यायी

57. विशाखदत्तः.

कुमारसम्भवम्

58. महर्षि पाणिनि

मुद्राराक्षसम्

भगवद्गुणगीतम्

पदानि शुद्धं कृत्वा गद्यखण्डं पुनः लिखत। (3)

59. अर्जुनः वृत्तांतं एतं श्रुत्वा पुरातनीयस्य महाभारताभिधस्य युद्धस्य परिणामम् स्मृतिपथमानयति । स विचारयति यत् तदा बहूनां जनानां संहारः संजातः, तेन च परिवारेषु आपदानां परंपरा प्रवर्तिता।

श्लोकस्य अनुवादं कृत्वा अर्थविस्तारं कुरुत। (3)

60. यथा चतुर्भिः कनकं परीक्ष्यते निघर्षणच्छेदनतापताडनैः।

तथा चतुर्भिः पुरुषः परीक्ष्यते श्रुतेन शीलेन गुणेन कर्मणा॥

61. सुखं हि दुःखान्यनुभूय शोभते घनान्धकारेष्विव दीपदर्शनम् ।

सुखात्तु यो याति नरो दरिद्रतां धृतः शरीरेण मृतः स जीवति ॥

-----|| इति शम् ||-----

Std: 10th

Date :29 / 01 /'25

Sub : Science and Technology

Total marks : 80

Instructions:

- 1) Write in a clear and legible handwriting
- 2) This question paper has four sections A,B,C & D and question Numbers from 1 to 54
- 3) All sections are compulsory. Internal options are given.
- 4) The numbers to the right represents the marks of the question
- 5) Draw neat diagrams wherever necessary
- 6) New sections should be written in a new page.

SECTION -A

→ Answer the following Q. no. 1 to 24 as directed: [1 mark each] (24)

- Choose the correct options :

1. What is the chemical formula of acetone ?

- (a) CH_3COOH
- (b) CH_3COCH_3
- (c) CH_3OH
- (d) HCOOH

2. What is the ratio of concentrated HCl and concentrated HNO₃ in aqua regia ?
- (a) 3 : 1
(b) 1: 3
(c) 2 : 1
(d) 1 : 2
3. How much charge is constituted by 18.75×10^{18} electrons ?
- (a) 1 C
(b) 2 C
(c) 3 C
(d) 0.5 C
4. An electric iron of power 2 kW is used for 3 hours. At Rs 5 per unit , the electricity bill will be Rs
- (a) 45
(b) 30
(c) 15
(d) 10
5. From the given table , find the correct option

Acid	Base	Nature of aqueous solution of salt produced			
		X	Y	Z	W
Strong	Strong	Neutral or slightly acidic or	Neutral	Acidic	Neutral or slightly acidic or basic
Strong	Weak	Basic	Acidic	Neutral or slightly acidic or basic	Basic
Weak	Strong	Acidic	Basic	Neutral	Acidic
Weak	Weak	Neutral	Neutral or slightly acidic or basic	Basic	Neutral

- (a) X
- (b) W
- (c) Z
- (d) Y

6. Which of the following group contains only bio-degradable substances ?

- (i) Grass , flower leather
- (ii) Grass , wood , plastic
- (iii) Fruit peels , cake , lemon juice
- (iv) Cake , wood , grass

- (a) (i) and (ii)
- (b) (i) , (ii) and (iv)
- (c) (i) , (iii) and (iv)
- (d) (ii) and (iv)

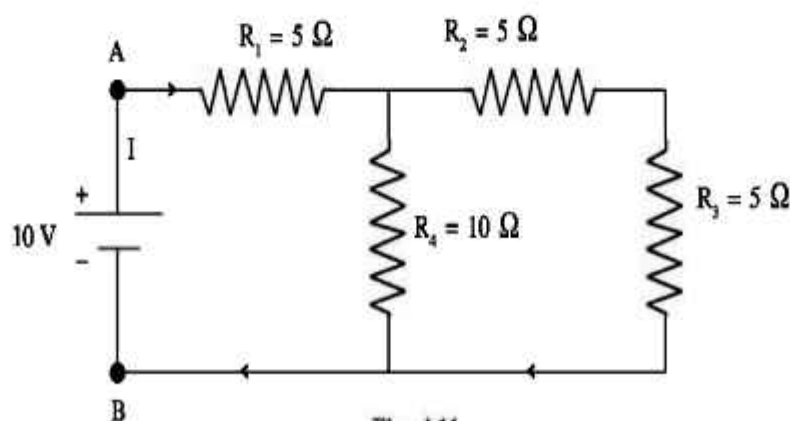
• Fill in the blanks to make the statements true

7. The molecular mass of cyclopentane is.....

8. The power of a convex lens is +4.0 D. The object should be placed at a distance ofcm to obtain the value of magnification -1.

9. Plasmodium divides into many daughter cells by

10. The equivalent resistance in the following circuit is Ω



11. The uterus opens into the vagina through

12. Two viral STD are and

- **Mention the below given statements are true or false**

13. $\text{pH} + \text{pOH} = 14$.

14. Any defect in optical nervous system or injuries to the brain can affect the vision even if the eyes are normal.

15. A clock when viewed in a plane mirror shows 04:30 . The correct time is 06 : 30.

16. The number of organisms are limited at each trophic level due to insufficient food supply.

- **Answer the following questions in short:**

17. What is the formula of sodium zincate ?

18. Mention the name of any one anti-oxidant gas.

19. A concave mirror produces a real image of an object placed at a distance of 10 cm from the mirror. If the image is enlarged by a factor of four relative to object then what is the focal length of the lens ?

20. Define : Angle of incidence

21. Mention the probability of birth of an offspring as a son or a daughter.

22. Two resistors of 6Ω and 4Ω are connected in parallel with a battery of V volts. The current passing in the circuit is 1.2 A then what is the current passing through a resistor of 6Ω ?

23. What is biomagnification ?

24. Mention any two terrestrial ecosystem.

SECTION -B

→ Question No. 25 to 37 each carries 2 marks. Answer in short in 40 to 50 words approximately. Attempt any 9 questions only (18)

25. Write chemical equation and balance them

- a) Potassium react with water to give potassium hydroxide and hydrogen gas
- b) Hydrogen sulphide gas burns in air to give water and sulphur dioxide

26. Answer the following :

- a) Why do acids not show acidic behaviour in absence of water ?
- b) Write the equation to show the reaction between plaster of paris and water

27. Answer the following :

- a) What happens to the image distance in the eye when we increase the distance of object from the eye ?
- b) Why does sky appears blue to us but dark instead of blue to an astronaut ?

28. Answer the following :

- a) How will you connect three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ to obtain total resistance of $1\ \Omega$?
- b) Why the conductors of electric heating appliances such as bread toasters and electric irons are made of an alloy rather than a pure metal ?

29. Mention the disorder shown in the figure and explain them in detail



30. Answer the following :

- Name the mirror used in the headlights of a car. Support your answer.
- Find the focal length of a lens of power -2.0 D. What type of lens is this ?

31. Name and explain two safety measures commonly used in electric circuits and appliances

32. Describe addition reaction of hydrocarbons in detail with proper example.

33. What are the methods used by plants to get rid of excretory products ?

34. Why is vegetative propagation practised for growing some type of plants ?

35. Samples of four metals A, B, C and D are added to following solution one by one. The results obtained are tabulated as follows:

Metals	Iron(ii) sulphate	Copper (ii) sulphate	Zinc sulphate	Silver nitrate
A	No reaction	Displacement		
B	Displacement		No reaction	
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

Observe the above table and answer the following questions

- Which is the most reactive metal ?
- Arrange the metals A , B ,C and D in the decreasing order of reactivity

36. Answer the following :

- Mention two ways how biodegradable substances affect the environment
- What is the role of seminal vesicles and prostatic gland ?

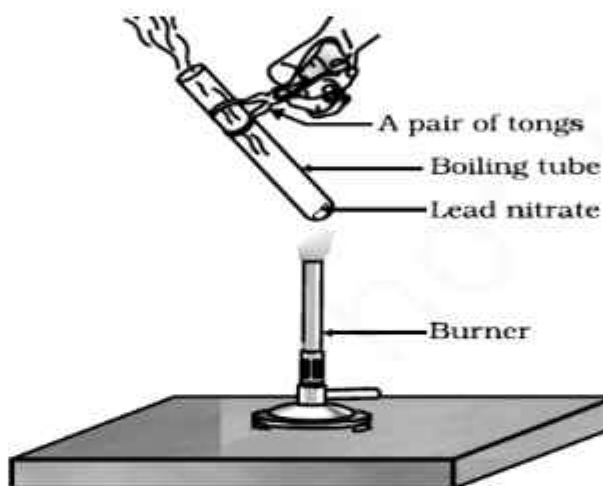
37. Describe method to obtain metals in the high of reactivity series in brief.

SECTION -C

→ Question No. 38 to 46 each carries 3 marks. Answer in short in 60 to 80 words approximately . Attempt any 6 questions only (18)

38. Observe the following diagram and answer the given questions.

- State the type of chemical reaction and define it
- Which gas is evolved from boiling tube ?
- Write the balanced equation of reaction.



39. Answer the following :

- Describe the importance of pH for the process of digestion
- Write formula and electron dot structure sodium oxide

40. Answer the following questions :

- Draw a diagram to show how hypermetropic eye is corrected .
- The near point of a hypermetropic eye is 1 m. What is the power of lens required to correct this defect ?

41. Explain in detail the process of sex determination in human being.

42. Explain significance of pancreatic hormone and disorders due to its irregular secretion. Also state some remedial measures.
43. Explain with proper examples.
- a) Combustion reaction of carbon compounds
 - b) Substitution reaction of carbon compounds
44. An object of size 7.0 cm is placed at a distance of 27 cm in front of concave mirror of focal length 18 cm. at what distance from the mirror should a screen be placed so that a sharp focussed image can be obtained ? Find the size and the nature of the image formed.
45. Explain structure of female reproductive system with diagram . Also explain the process of menstruation
46. Describe parallel connection of resistors and derive the equation for equivalent resistance

SECTION -D

→ **Question No. 47 to 54 each carries 4 marks Answer in long in 90 to 120 words approximately. Attempt any 5 questions only (20)**

47. Answer the following :
- a) Draw the molecule of soap and detergent.
 - b) Describe the cleansing action of soap and detergent.
48. Write a short note on human brain with a neat and labelled sketch
49. Describe refraction of light through rectangular glass slab. Define lateral shift.
50. Answer the following:
- a) Explain scattering of light
 - b) Why sky appears blue ?
-

- c) If earth has no atmosphere then how sky appears to us ? Why ?
- d) What is the reason for reddish appearance of sun during sunrise and sunset ?

51. With a neat and labelled sketch , explain human respiratory system and its function in detail.

52. Describe an activity to understand the reaction of metal carbonates and metal hydrogen carbonates with acid with necessary diagram and equation

53. Answer the following :

- a) Explain series connection of resistors and derive the equation for equivalent resistance.
- b) An electric heater of resistance $8\ \Omega$ draws 15 A current from the mains for 2 hours. Calculate the rate at which heat is developed in the heater.

54. Answer the following :

- a) Draw neat and labelled sketch of domestic electric circuit.
- b) Explain overloading and short circuit
- c) What is the function and importance of earth wire

SOCIAL SCIENCE

Std:-10th Eng.Med.

Time:3 Hours
Total Marks:-80

SECTION-A

Answer the following questions as required(Each question carries 1 Mark).

- Match the following correctly:

1) Muslin of Dhaka	a) Deforestation
2) King Ashoka	b) Privaization
3) Destruction of forest	c) Globalization
4) Solar Energy Plant	d) Match Box
5) Evil of monopoly incaused	e) Madhopur
	f) Five Stupas

- State whether the following statements are True or False:

6. Upnishad literature is in dialogue form.
7. Minakshi temple has main five Gopuram.
8. Vallabhi University was the centre on Hinyan Buddhist Cult.
9. Ralph Nadar is known as the pioneer of consumer movement.
10. India is the largest democracy in the world.

- Fill in the blanks with correct options from those given in the brackets:

11. The Hemchandracharya Library is located at _____. (Ahmedabad, Gandhinagar, Patan)
12. Shri Mahavir Jain Aradhana Kendra Koba is located at _____. (Patan, Ahmedabad, Gandhinagar)
13. Apart from agriculture there is maximum use of water in _____. (Cleaning, Household, Industries)
14. In Mizoram _____ % Area of total sown area is under irrigation. (12.5, 7.3, 15.4)
15. A dense network of railway is seen in Ganga Plain due to _____ and dense population. (canals, industries, intensive agriculture)

▪ **Choose the correct option from those given below and answer the question:**

16. Who had addressed the World Religious Conference held at Chichago.

- (A) Swami Ramkrisna Paramhansa (C) Raja Ram Mohan Roy
(B) Swami Dayanand Saraswati (D) Swami Vivekanand

17. On which river is the Grand Anicut Canal (Grand Dam) constructed?

- (A) Godavari (B) Kaveri (C) Krishna (D) Tungbhadra

18. After Independenc, by how many times has irrigated area increased in India?

- (A) One and half times (C) Three times
(B) Two and half times (D) Four times

19. Which sector is dominant sector in the developing countries ?

- (A) Commercial (B) Primary (C) Service (D) Secondary

20. Which river in Odisha has a delta.

- (A) Krishna (B) Mahanadi (C) Kaveri (D) Mahi

▪ **Answer the following in short:**

21. Which is the natural factor of production ?

22. What are the two main methods of allocation of means of production ?

23. In which mountaineous area of South India is a ropeway installed ?

24. Which is the biggest port of Gujarat?

SECTION-B

Answer any 9 out of 13 questions given below in 30 to 35 words:(Each question carries 2 marks each)

25. Explain the types of forests according to administration.
26. State the use of Manganese.
27. Write short note on Bio-Gas.
28. Identify me:- 1) I am known as "Silicon Valley" of India. 2) I am known as the "Sunrise Industry".
29. Mention the steps to curb environmental degradation.
30. Write a note on Naxalite movement.
31. The Right to Freedom of Religion assures the Minorities of their rights. Explain the statement.
32. Give information about the most ancient people belonging to the Dravidian race.
33. Which are the things included in India's natural heritage ?
34. Name the metal items produced by the artisans of Lothal ?
35. "The relationship of human life has been connected with the tree and the forest since the beginning".
Explain the statement .
36. Give information about Qutub Minar.
37. Write a brief note on Ellora caves.

SECTION-C

Answer any 6 out of 9 questions given below in 60 to 80 words:(Each question carries 3 marks)

38. Give information about Harappa as an ancient town of India.
39. Explain town-planning in ancient India.
40. Write a note on Takshshila University.
41. Describe the process of soil formation and state the basis on which these are classified.
42. Explain the following terms: 1) Regur soil, 2) Marshy or Peaty soil, 3) Desert soil.
43. Explain the advantages and disadvantages of Globalization.
44. "Price rise is beneficial to as well as a hindrance in economic development". Explain the statement .
45. Explain the role of Public distribution system in controlling price rise.
46. What is Women empowerment? What are the obstacles / difficulties in its path?

SECTION-D

Answer any 4 questions out of 7 questions given below in 120 to 150 words. Question no. 54 is compulsory. (Each carries 4 marks)

47. "Chemistry is an experimental science" Explain the statement.
48. Explain the importance of ancient India in the field of Medicine and Surgery.
49. What are the types of farming in India? Explain them in detail.
50. State the technical reforms brought in the field of Indian agriculture.
51. Write names of any five schemes/ programmes undertaken by government for poverty eradication and explain the given below shemes: 1) MAA Annapurna Yojana, 2) Encouragement to organic farming.
52. Write the meaning of unemployment and describe its various types in detail.
53. Mention the 'Juvenile Rights' as per the Constitution.
54. Show following details in outlined map of India with proper symbols at proper place.
 - (1) Black soil
 - (2) Periyar Sanctuary
 - (3) One state growing groundnut
 - (4) Ship building industry

..BESTOFLUCK..

SOCIAL SCIENCE

Std:-10th Eng.Med.

Time:3 Hours
Total Marks:-80

SECTION-A

Answer the following questions as required(Each question carries 1 Mark).

- Match the following correctly:

1) Tiger and peacock	a) Jamnagar
2) Dance of Padhaar tribe	b) Manipur Tamil project
3) Forest soil	c) Padhaar dance
4) Specific deer species	d) Jaisalmer
5) Reserve of natural gas	e) Terai region of Central Himalaya
	f) Vehicle of Deities

- State whether the following statements are True or False:

6. The most ancient language of Dravidian family is Telugu.
7. The microbes make humus and thus make nutrients available to the roots.
8. Mean years of schooling are obtained as the years spent in the school by an adult till the age of 25.
9. A national minority commission has been set up for protecting the rights, interest, welfare and for progress of minorities.
10. At an junction or roundabout vehicle coming from the right should be given priority.

- Fill in the blanks with correct options from those given in the brackets:

11. gave the concept of poverty line first. (WTO, WHO, WNO)
12. There are/is bio-reserve zone in Gujarat. (1, 10, 2)
13. Irrigation is practiced in about of net sown area. (48 %, 38%, 28 %)
14. In Punjab% area of the total area sown is under irrigation. (40%, 90.8 %, 28 %)
15. India has about km long coastline. (7515, 6516, 7516)

- Choose the correct option from those given below and answer the question:

16. Geometrical design shown in image is situated in which city ?



- (A) Ahmedabad (B) Patan (C) Junagadh (D) Vadnagar

17. Select the correct option from the given options considering the following pairs.

1. Bhakhra Nangal	a. Madhya Pradesh, Gujarat, Rajasthan, Maharashtra
2. Narmada valley	b. Karnataka and Tamil Nadu
3. Tungabhadra	c. Andhra Pradesh, Telangana
4. Nagarjunsagar	d. Punjab, Haryana, Rajasthan
	e. Karnataka, Andhra Pradesh

- (A) (1-b), (2-a), (3-d), (4-e) (C) (1-c), (2-e), (3-b), (4-d)
(B) (1-d), (2-c), (3-b), (4-e) (D) (1-d), (2-a), (3-e), (4-c)

18. Select the correct option from the given options considering following options?

1. The law of who tills the field	a. Green revolution
2. Land Ceiling Act	b. Pradhanmantri fasal bima yojna
3. Insurance protection to agricultural crops	c. Path revolution
4. To increase the agricultural yield	d. The disparity among the land owners has been removed
	e. Land to the tiller

- (A) (1-d), (2-e), (3-a), (4-c) (C) (1-c), (2-d), (3-b), (4-e)
(B) (1-e), (2-c), (3-d), (4-a) (D) (1-e), (2-d), (3-b), (4-a)

19. Match the following :

1. Capitalist system	a. Natural factor of production
2. Primary sector	b. Aiming to get maximum income and expenditure
3. Land	c. Factors of production owned by the private
4. Economic activities	d. Inclusion of agro activities

- (A) (1-a), (2-b), (3-c), (4-d) (C) (1-b), (2-c), (3-d), (4-a)
(B) (1-c), (2-d), (3-a), (4-b) (D) (1-c), (2-b), (3-a), (4-d)

20. Which vehicles can not be driven in a two - wheeler lane ?

- (A) Motor car (C) (A) and (B) both
(B) Heavy vehicles (D) Gear less motor cycle

▪ **Answer the following in short:**

21. Dr. Subhash Palekar has classified / divided nature's arrangement into which four types ?
22. Which are the things included in living standard?
23. Which are important ports of Gujarat ?
24. Airport Authority of India carries out management of which airport?

SECTION-B

Answer any 9 out of 13 questions given below in 30 to 35 words:(Each question carries 2 marks each).

25. What do you know about places of Gujarat having historical importance ?
26. Give Reasons : Since ancient times a composite culture has been created in India.
27. Write a short note on Jari work of India.
28. Give introduction to wooden art of ancient India.
29. What is national emblem ?
30. Write a note on temple architecture.
31. Give introduction to India's wildlife diversity.
32. Explain statements/reasons : Coal is a multiuse/multipurpose resource.
33. Write properties and uses of iron.
34. Write a short note on sugar industry.
35. What do you know about plastic industry ?
36. Why there are serious challenges raised against social harmony, secularism, democratic values and national integrity ?
37. Give reason :- In India the Government has set up national minority commission.

SECTION-C

Answer any 6 out of 9 questions given below in 60 to 80 words:(Each question carries 3 marks)

38. Write a short note on Medieval Architecture.
39. Write a note on town planning of city Harappa.
40. Write a short note on Varanasi education centre.
41. 'Resources are useful in different ways.' Explain.
42. 'Alluvial soil is attributed to the deposition of alluvium'. Explain.
43. Give reason : Globalization leaves dual effects on Indian economy.
44. Explain following statement/reason : Price rise leads the society to moral destruction.
45. Write a short note : Increase in money supply.
46. What is meant by women empowerment ? Why is women empowerment needed in India ?

SECTION-D

Answer any 4 questions out of 7 questions given below in 120 to 150 words Question no. 54 is compulsory. (Each carries 4 marks)

47. Write a short note on Vastushastra.
48. Show the importance of science and technology.
49. What is meant by cash crops ? Give explanation about main cash crops of India.
50. Which are main agriculture crop's main producing states of India ? Evaluate it.
51. Explain unemployment and discuss its types.
52. Write a short note on labour resource planning.
53. The picture shown below is of a worldwide problem. What is that problem ? Give information about it



54. Show following details in outlined map of India with proper symbols at proper place.

- | | |
|--------------------------------|----------------------------------|
| (1) Bandipur – national park | (3) One region producing uranium |
| (2) One region producing cocoa | (4) Tapi river |

..BEST OF LUCK..

SOCIAL SCIENCE

Std:-10th Eng.Med.

Time:3 Hours
Total Marks:-80

SECTION-A

Answer the following questions as required(Each question carries 1 Mark).

- Match the following correctly:

1) Jayshankar Sundari	a) India
2) Malav lake	b) dramatic art
3) Desert soil	c) Viramgam
4) Henotaro	d) Some areas of Kachchh and Saurashtra
5) Fifth country getting wind energy	e) Banni area of Kachchh district
	f) China

- State whether the following statements are True or False:

6. To determine health and longevity, life expectancy is considered at the time of death of an infant.
7. Rebellion arises from local dissatisfaction while terrorism has no restrictions.
8. Natural arrangement is such that whatever elements came from wherever they have to return there.
9. Do not honk unnecessarily or continuously or in a designated quiet areas while driving.
10. Awadhi language is known as the language of Aryans or sages.

- Fill in the blanks with correct options from those given in the brackets:

11. The market is called where the nations undertake exchange of labourers. (world labour, world employment, world unemployment)
12. There are about species of animals-birds in the world. (10 lakh, 15 lakh, 20 lakh)
13. multipurpose project is located on river Mahisagar. (Hirakubnd, Bhakara-nangal, Kadana-Vanakbori)
14. Availability of pure potable water is the need of life. (common, basic, different)
15. Four metro cities Delhi, Mumbai, Chennai and Kolkata will be interlinked by system. (Shipping Coporation of India, Golden Quadrilateral Highway, Multipurpose System)

- Choose the correct option from those given below and answer the question:

16. Given sculpture image is found in which cave ?



- (A) Ajanta (B) Ellora (C) Elephanta (D) None of this

17. Select the correct option from the given options considering the following pairs.

1. Watershed development	a. Main medium of irrigation.
2. Rain water harvesting	b. Water conservation
3. Maintenance of water resource	c. Main medium of irrigation.
4. Well and tubewell	d. Basin region of branch river.
	e. To increase the capacity of ground water.

- (A) (1-c), (2-a), (3-e), (4-b) (C) (1-d), (2-c), (3-b), (4-e)
 (B) (1-b), (2-e), (3-a), (4-c) (D) (1-d), (2-e), (3-b), (4-a)

18. Crops and other farm products hold aboutcontribution in export also which earns huge foreign exchange.

- (A) 17% (B) 22% (C) 18% (D) 60%

19. Which of the following is incorrect in terms of advantages of Market mechanism system ?

- (A) Maximum production can be attained
 (B) New research takes place which accelerates the economic development
 (C) Maximum expenditure
 (D) Best Quality of commodity due to competition

20. A person aged 16 years or older can get a license to drive a two wheeler without gear motorcycle with an engine up to what capacity ?

- (A) 50 (B) 100 (C) 120 (D) 150

- Answer the following in short:

21. At which temperature do local earthworms continue their activities under the soil ?
 22. Which constructive change has been done in communication system ?
 23. Which cultural factors affect the transportation ?
 24. For whom are the material facilities or comfort but only an imaginary matter ?

SECTION-B

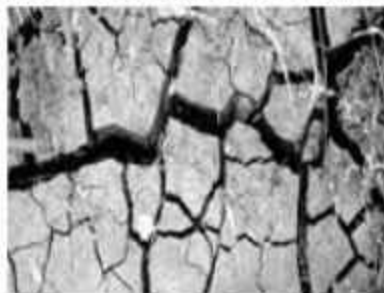
- **Answer any 9 out of 13 questions given below in 30 to 35 words:(Each question carries 2 marks each).**

25. The importance of Himalaya mountain is high for India ? Why ?
26. Write a short note on Mongoloid tribes.
27. Write a short note on art of dance.
28. Mention contribution of scholars in Sanskrit literature.
29. Give information about construction of houses of Mohan-Jo-Daro.
30. Give information about linear sketch of stupa.
31. Today the existence of wild life and birds is in danger. Explain the statement.
32. Give properties and usages of aluminium.
33. Explain statements/reasons : Iron is considered to be the base of industrial development of modern world.
34. Give information about synthetic cloth.
35. Review paper industry as agro based industry.
36. Why there are serious challenges raised against social harmony, secularism, democratic values and national integrity ?
37. Give reason :- Communal ideology leads society towards division.

SECTION-C

- **Answer any 6 out of 9 questions given below in 60 to 80 words:(Each question carries 3 marks)**

38. Chariot temples are considered to be the immortal heritage of art of architecture - Explain the statement.
39. Give information about linear sketch of temple having Gopuram architecture.
40. Evaluate Amir Khusro as Persian literature artist of medieval age.
41. Lava rocks type soil in this picture ? Write notes on it.



42. Write a note on the types of resources.
43. Write a short note on liberalization policy in India.
44. Write a note on price control system.
45. Write a note on consumer protection.
46. Write a short note on progress in the field of health in India.

SECTION-D

- **Answer any 4 questions out of 7 questions given below in 120 to 150 words Question no. 54 is compulsory. (Each carries 4 marks)**

47. Write a note on discoveries made by Aryabhatta in mathematics.
48. Ancient Nalanda University is known as chemistry lab. Explain.
49. Mention main/major agricultural products of India.
50. Write a short note on Green revolution.
51. Explain unemployment and discuss its types.
52. Which solutions are undertaken by government for setting unequal distribution of income ?
53. Describing efforts to remove child labour from India and explain about their welfare schemes.
54. **Show following details in outlined map of India with proper symbols at proper place.**
 - (1) One centre of rope-way
 - (2) One region producing tea
 - (3) Dharoi project
 - (4) One centre of cotton cloth industry

..BEST OF LUCK..